

Strategic Communication Under Perceived Misalignment of Interests

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Abstract. Why might Senders fail to persuade when they know their interests align with a Receiver's, but the Receiver is uncertain about that alignment? I study this question using a novel communication game in which a Sender maximizes their and a Receiver's pay-offs by distorting private information. In an experiment that distinguishes several mechanisms for communication failures, I find that Senders frequently over-communicate. Over-communication is primarily driven by a failure to acknowledge Receiver skepticism, rather than incorrect beliefs about the Receiver's strategy. These findings highlight a strategic reasoning error that can arise even when incentives align and strategic beliefs are correct.

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1 Introduction

In many settings where an informed *Sender* communicates with an uninformed *Receiver*, the Sender knows their interests align with the Receiver’s, while the Receiver is unsure of this alignment. For instance, financial advisors are widely viewed as untrustworthy (Egan et al., 2024), even though many have a fiduciary duty to their clients. Decision-makers also frequently reject recommendations from advisors seeking to help them achieve financial and health-related goals, especially when they lack trust in the advisor (Bhattacharya et al., 2012; Cole et al., 2013; Burke and Hung, 2021; Alsan and Eichmeyer, 2024; Banerjee et al., 2024). Finally, “zero-sum thinking” is prevalent in political decision-making: many voters believe certain policies will benefit others at their own expense, while those policies actually benefit the average voter (Ali et al., 2025; Chinoy et al., 2026).

In such settings, successful persuasion allows the Sender to induce actions that are beneficial to both them and the Receiver. Such an outcome is only achievable, however, if the Sender communicates in a way that addresses the Receiver’s skepticism about the Sender’s incentives. Specifically, successful persuasion may require the Sender to distort their private information, not because their preferences conflict with the Receiver’s, but because the Receiver fears that they do.

This paper studies why Senders may fail to implement this strategic insight in practice. That is, why might a Sender fail to communicate persuasively when they know their interests align with a Receiver’s, but the Receiver is unsure of this alignment? When communicating parties’ preferences are known to be misaligned, deviations from theoretically-optimal communication are often driven by a *belief error*: Senders hold incorrect beliefs about the Receiver’s response (Blume et al., 2020; Lafky et al., 2022). This paper studies a setting in which Senders may additionally display a *reasoning error*: knowing that incentives are aligned, they may not recognize the need to account for the Receiver’s potentially skeptical response. People struggle to reason from the perspective of others who are less informed (Camerer et al., 1989) and to condition on relevant strategic contingencies (Niederle and Vespa, 2023), yet it is unclear whether such errors impair communication. Distinguishing belief errors from reasoning errors is important because they call for different interventions. Communication failures driven by incorrect beliefs may be corrected by feedback about the Receiver’s response to recommendations, while failures driven by reasoning errors require helping Senders recognize and account for the Receiver’s skepticism.

Identifying the source of such communication failures requires a design that separates observationally similar mechanisms. A Sender who reveals too much information may be best-responding to an incorrect belief that Receivers are credulous (a belief error), or may anticipate

the Receiver’s skepticism but fail to account for it when communicating (a reasoning error). I address this identification problem by introducing a *recommendation game* with a sharp benchmark for optimal communication to a wary Receiver, and by implementing a laboratory experiment that disentangles belief and reasoning errors.

In the recommendation game, a Receiver can either accept a Sender’s recommended action or reject it in favor of a status quo action. The Receiver wants their action to be as close as possible to the state-matching action, and is unsure whether the Sender wants the same thing. Specifically, there are two types of Sender: an *aligned type* knows the state and has the same preferences as the Receiver, while a *commitment type* always recommends the highest action. When the Receiver has a sufficiently low expected payoff from taking the highest action, the game has a unique pure-strategy *wary equilibrium* and no mixed-strategy equilibrium. In a wary equilibrium, the aligned-type Sender must distort their private information to improve their and the Receiver’s payoff relative to the status quo. This is because, given the Receiver’s low payoff to taking the highest action, it is optimal for them to accept all recommendations except the recommendation to take the highest action (i.e., play an *accept-reject* strategy). Anticipating this response, the aligned-type Sender should play a *partial truth-telling* strategy: recommend the state-matching action in all states except the highest one, where they should recommend a more moderate action.

Partial truth-telling is thus the theoretical benchmark for optimal communication to wary Receivers in the recommendation game. Having pinned down this benchmark, I turn to the lab to evaluate why people acting as aligned-type Senders may fail to use this strategy when communicating to Receivers who fear a misalignment of interests. To that end, in the experiment, the recommendation game is parametrized such that the wary equilibrium is its unique equilibrium. Participants play the game as either a Receiver or an aligned-type Sender; the commitment-type Sender is played by a computer. Accordingly, I use the term “Sender” to describe a participant in the aligned-type Sender role.

In the *main treatment*, participants play twenty rounds of the recommendation game. After each round, they receive feedback about their opponent’s strategy and are randomly re-matched to a new opponent. Importantly, this feedback informs Senders how a Receiver would have responded to each recommendation that involves switching from the status quo, allowing them to update potentially incorrect beliefs about the Receiver’s strategy.

The main treatment’s key finding is that Senders systematically communicate suboptimally. First, I establish that most Receivers play their theoretically-optimal accept-reject strategy. This makes partial truth-telling empirically optimal for Senders, and the feedback Senders receive allows them to learn this. Nevertheless, only 21% of Senders play partial truth-telling in the first round. Instead, nearly two-thirds of them always recommend the action corresponding

to the state. That is, most Senders send recommendations that would be optimal if Receivers accepted all recommendations.

Given this systematic deviation from optimal communication, I next investigate the source of Senders' error. I focus first on a potential belief error: consistent with level- k reasoning (Stahl and Wilson, 1994; Nagel, 1995), Senders may be best-responding to the belief that Receivers accept all recommendations. If this explanation is correct, Senders should become more likely to play partial truth-telling if they receive more feedback that a Receiver has played accept-reject. However, feedback has at best a limited effect on Senders' strategies. In the twentieth and final round, when the average Sender has been paired with fourteen Receivers who played accept-reject, fewer than half of Senders (45%) play partial truth-telling. While this is a significant increase relative to the first round, regression analyses show that the effect cannot be linked to the receipt of feedback that a Receiver played accept-reject. Incorrect beliefs therefore do not appear to be the primary cause of Senders' communication failure.

Another possibility is that Senders display a reasoning error: they fail to realize that they should not recommend the best possible action for the Receiver, but rather the best possible action that the Receiver will accept. This error could be captured by a *strategic myopia*-driven *failure of contingent thinking*: a failure to best-respond to a player's strategy unless that strategy is made salient (Niederle and Vespa, 2023). In guessing games, for instance, this type of failure of contingent thinking limits players' ability to best-respond to their own past action unless they are first prompted to replicate it (Fragiadakis et al., 2016). Thus, if Senders' initial mistake reflects a failure of contingent thinking, making the Receiver's accept-reject strategy more salient should increase Senders' likelihood of playing partial truth-telling. To evaluate this possibility, Senders play an additional round of the recommendation game where they directly respond to the Receiver's accept-reject strategy.

I find that this intervention makes Senders much more likely to play partial truth-telling. Whereas only 45% use this strategy in the final round, 78% do so when responding directly to the Receiver's accept-reject strategy in the additional round. The increase occurs primarily because many Senders who did not play partial truth-telling in the final round switch to partial truth-telling in the additional round. Senders' behavior in the additional round thus provides evidence in favor of a failure of contingent thinking. It also provides evidence against two alternative mechanisms: a strict preference for truth-telling, which would predict full truth-telling regardless of the Receiver's response, or an inability to identify the best response to accept-reject.

I conduct a second treatment, the *beliefs treatment*, to collect additional evidence on the mechanism driving Senders' communication failure. Before selected rounds, Senders report

their beliefs about the modal Receiver strategy in the upcoming round.¹ This design serves two purposes. First, it provides a direct test of the incorrect-beliefs mechanism by measuring whether Senders correctly predict that Receivers tend to play accept-reject. Second, it functions as a salience intervention: if Senders report that the modal Receiver plays accept-reject, then the elicitation task directs their attention to the Receiver strategy they should respond to. Thus, under a failure of contingent thinking, Senders in the beliefs treatment should be more likely to play partial truth-telling than Senders in the main treatment if they correctly identify accept-reject as the modal Receiver strategy.

The beliefs treatment provides further support for the failure of contingent thinking mechanism. Most Senders correctly state that the modal Receiver plays accept-reject, indicating that the belief elicitation task successfully directs attention to the relevant contingency. The prevalence of these correct beliefs also further weakens the incorrect-beliefs explanation. The elicitation task also affects Senders' strategies: consistent with a failure of contingent thinking, Senders are substantially more likely to play partial truth-telling in the beliefs treatment. In each round, the share of Senders using this strategy is up to 84 percent higher than in the corresponding round of the main treatment. Moreover, Senders with correct beliefs about the modal Receiver strategy - that is, those who focus on the relevant contingency - are consistently more likely to play partial truth-telling.

Taken together, these results support a failure of contingent thinking: many Senders communicate suboptimally because they fail to focus on how the Receiver may respond. As a result, the most effective remedies for this communication failure involve making the Receiver's strategy salient *ex ante*, as opposed to providing *ex-post* feedback about it. The cause of the failure is therefore distinct from some of the most well-understood strategic reasoning errors, such as mispredicting an opponent's strategy or being unable to compute one's best response (Eyster, 2019). A key implication is that strategic reasoning errors may arise even when players' incentives are aligned and their beliefs about others' strategies are correct.

The results also have implications for paternalistic interactions, in which informed agents try to improve others' choices. The Sender's recommendation can be viewed as a "soft" paternalistic intervention: one that preserves the Receiver's freedom of choice. Earlier work suggests that both paternalistic agents and decision-makers prefer soft interventions over "hard" ones that restrict decision-makers' choice sets (Fehr et al., 2013; Bartling et al., 2014, 2025).² How-

¹It is unclear how eliciting beliefs about opponents' strategies affects players' own strategies. While some studies find that belief elicitation affects players' strategies (Erev et al., 1993; Croson, 1999, 2000; Rutström and Wilcox, 2009; Gächter and Renner, 2010), others find no such effect (Nyarko and Schotter, 2002; Costa-Gomes and Weizsäcker, 2008).

²When paternalists do use hard interventions, their goal is often to align decision-makers' choices with their own aspirations (Ambuehl et al., 2021).

ever, precisely because they preserve choice freedom, soft interventions are only effective if they are designed with the decision-maker’s response in mind. My findings suggest that even well-intentioned paternalistic agents may fail to design effective soft interventions, but point to decision aids that may help them do so.

Connections to Related Literature. I focus on a form of communication that has not been studied in prior experiments: players’ preferences are fully aligned, but strategic tension arises because Receivers are uncertain about that alignment. In contrast, prior work has focused on settings where players’ preferences are misaligned, and the degree of misalignment is common knowledge (Dickhaut et al., 1995; Blume et al., 1998, 2001; Kawagoe and Takizawa, 2009).³ While Senders tend to over-communicate in these games - that is, reveal more information than the theoretical benchmark (Cai and Wang, 2006; Wang et al., 2010; Lafky et al., 2022) - this over-communication is typically driven by incorrect beliefs about the Receiver’s strategy (Blume et al., 2020; Lafky et al., 2022). The present study shows that, in a setting with a perceived rather than true misalignment of interests, over-communication stems from a reasoning error rather than a belief error. These findings indicate that there are multiple possible causes of over-communication, and point to simple interventions that mitigate the form that is driven by a failure of contingent thinking.

The recommendation game is conceptually similar to a noisy cheap talk game: a variant of canonical cheap talk (Crawford and Sobel, 1982) in which the Receiver is transmitted a noisy signal of the Sender’s original message (Myerson, 1991; Blume et al., 2007; Lightle, 2014; Blume et al., 2023). In both games, the Sender’s optimal strategy accounts for the Receiver’s uncertainty about the original message. In the recommendation game, however, this uncertainty stems from the two possible Sender types rather than noisy communication by a single type of Sender, which facilitates extensions to settings where the misaligned Sender is strategic (see Section 5).⁴ Moreover, the focus of noisy cheap talk is fundamentally different: while it studies how noise may improve information transmission by a Sender with misaligned interests, the recommendation game features a Sender with aligned interests who must account for the Receiver’s fear that interests may be misaligned. The recommendation game also resembles political correctness and bad reputation models, where a “good” Sender distorts their private information or acts myopically out of concern for their reputation (Sobel, 1985; Benabou and Laroque, 1992; Morris, 2001; Ely and Välimäki, 2003; Ely et al., 2008). The wary equilibrium

³Blume et al. (1998, 2002) study strategic communication in games where the Sender and Receiver share a common interest. In their games, however, messages have no meaning a priori; the studies focus on players’ ability to develop a commonly-understood language.

⁴Most cheap talk games feature a Sender with a single preference type. Li and Madarász (2008) study a cheap talk game where the Sender has two possible preference types, but unlike in the recommendation game, both types’ preferences differ from the Receiver’s.

of the recommendation game points to a force beyond reputational concerns that may lead “good” Senders to distort their information.

This paper also builds upon earlier work on reasoning errors. While failures of contingent thinking are known to limit people’s ability to extract information from others’ strategies (Charness and Levin, 2009; Esponda and Vespa, 2014, 2024; Martínez-Marquina et al., 2019; Ali et al., 2021; Ngangoué and Weizsäcker, 2021) and best-respond to those strategies (Fragiadakis et al., 2016), my findings show that such errors can additionally impair people’s ability to communicate persuasively. Since Senders struggle to ignore that they are better-informed about their preferences than the Receiver is, their reasoning error is similar to a curse of knowledge (Camerer et al., 1989). Here, however, Senders are “cursed” with their knowledge of the alignment of their preferences with the Receiver’s, rather than the state of the world. Finally, the interventions that mitigate Senders’ reasoning error resemble those that mitigate correlation neglect, which involve redirecting attention toward a key structural feature of the decision problem (Enke and Zimmermann, 2019; Rees-Jones et al., 2024). This parallel suggests that, in many cases, people may not be intrinsically unable to overcome a reasoning error; they may simply need to be nudged to consider the information that is relevant to their decision.

The rest of the paper proceeds as follows. Section 2 characterizes a theoretical benchmark for optimal communication to wary Receivers, and discusses behavioral mechanisms that may prevent Senders from communicating optimally in practice. Sections 3 and 4 present the design and results of the main and beliefs treatments of the experiment. Section 5 concludes.

2 Optimal Communication Benchmark and Behavioral Predictions

This section develops the theoretical benchmark and behavioral predictions that guide the experiment. I first characterize optimal communication by a Sender whose interests are aligned with a Receiver’s, but who faces a Receiver who is uncertain about that alignment. I then formalize two behavioral mechanisms that may prevent a Sender from using that optimal communication strategy in practice. All proofs are in Appendix A.

The optimal communication benchmark is pinned down in a novel *recommendation game* played by a Sender and a Receiver. The Sender has private information about the payoff-relevant state, and the Receiver is uncertain whether the Sender’s interests align with their own. The central question is how a Sender whose preferences align with the Receiver’s - an *aligned-type* Sender - should communicate when this uncertainty leads the Receiver to distrust some recommendations. Intuitively, the aligned type’s optimal strategy involves communicating in

a way that addresses the Receiver’s skepticism about the Sender’s incentives.

The behavioral analysis then considers why aligned-type Senders may fail to implement the theoretically-optimal communication strategy. I focus on two mechanisms: incorrect beliefs about the Receiver’s strategy and a failure of contingent thinking. Each mechanism generates a distinct prediction about when aligned-type Senders should fail to communicate optimally, and the type of intervention that should correct this failure.

2.1 Primitives

There are three states, $\theta \in \{1, 2, 3\}$, and three corresponding actions, $a \in \{1, 2, 3\}$. Let $\hat{a}(\theta)$ denote the action corresponding to state θ . There are two players: a *Sender* and a *Receiver*. The Sender is one of two types. With probability p , the Sender is an *aligned type*: they observe the state and have the same utility function as the Receiver. With probability $1 - p$, the Sender is a non-strategic *commitment type*: they always recommend the same action, c .

The Sender’s aligned type represents an expert advisor whose interests coincide with a decision-maker’s, such as an experienced financial advisor with a fiduciary duty. The Sender’s commitment type can be interpreted either as an advisor with conflicting incentives, such as pressure to sell particular financial products,⁵ or as a well-meaning advisor who recommends products suited to the average client rather than tailoring their recommendations.

The Receiver does not observe the state. They have a prior over states that assigns probability $q_\theta > 0$ to state θ , where $\sum_\theta q_\theta = 1$. The prior (q_1, q_2, q_3) is independent of p , and all parameters are common knowledge. Let s denote the Receiver’s *status quo*, interpreted as an action the Receiver is already taking or one they believe is likely to correspond to the state. I assume $|s - c| = 2$, so the status quo and the commitment type’s recommendation are at opposite boundaries of the action space. The values of s and c are common knowledge. Without loss of generality, let $s = 1$ and $c = 3$.⁶

The Receiver’s payoff decreases in the distance between their action and the action corresponding to the state. Their preferences are described by $u(\alpha|a_R(r) - \hat{a}(\theta)|)$, where $\alpha > 0$ is a scaling parameter and $u'(\cdot) < 0$. An aligned-type Sender has the same utility function as the Receiver. I impose the following tie-breaking assumption, which is similar to the “truthful when convenient” assumption made in equilibrium refinements of other communication games (Fr chet te et al., 2022): when an aligned-type Sender is indifferent between two or

⁵Such pressure is common in the financial industry. For instance, financial advisors at major Canadian banks are frequently pressured, and sometimes even coached, to recommend certain products to their clients (Johnson, 2017; Johnson et al., 2024).

⁶The fact that 1 is the Receiver’s status quo is a behavioral assumption. That is, under the assumed structure of the Receiver’s prior, 1 is not necessarily the Receiver’s ex-ante optimal action. Lemma 2 (see Appendix A.3) pins down the condition on the Receiver’s prior that makes 1 ex-ante optimal.

more recommendations, they choose the recommendation closest to $\hat{a}(\theta)$.

Given these primitives, the game proceeds in three stages. First, Nature determines the state and the Sender's type. Second, the Sender recommends an action $r(\theta)$ to the Receiver. The commitment type recommends $r(\theta) = 3$ for all θ , while the aligned type observes the state and recommends some $r(\theta) \in \{1, 2, 3\}$. Third, the Receiver observes the recommendation r and either accepts or rejects it. Accepting entails taking the recommended action, while rejecting entails taking the status quo; that is, $a_R(r) \in \{1, r\}$.⁷ This action space captures situations in which the Sender makes actions available to the Receiver, either by providing the opportunity to take them or by reducing search costs. For instance, a client may hire a financial advisor to reduce the costs of identifying an ideal investment portfolio, and the advisor may enable the client to purchase customized financial products.

2.2 Equilibria and Robustness

The experiment focuses on a parameter region in which the Receiver's wariness about the Sender's incentives makes full truth-telling ineffective for the aligned-type Sender. In this region, the Receiver rationally rejects a recommendation of 3. The aligned-type Sender's best response is then to recommend 2 in state 3, and to recommend the action corresponding to the state in all other states. This *partial truth-telling* strategy is the optimal communication benchmark for the experiment.

To place this benchmark in context, Proposition 1 characterizes the game's two pure-strategy weak perfect Bayesian equilibria. Partial truth-telling is the aligned-type Sender's optimal strategy in a *wary equilibrium*. When the Receiver's expected payoff from accepting a recommendation of 3 is sufficiently high, the recommendation game also has a *trusting equilibrium* where the Receiver accepts all recommendations, allowing the aligned-type Sender to report the state truthfully.⁸

PROPOSITION 1: *The recommendation game has two pure-strategy weak perfect Bayesian equilibria. When $q_1 \geq q_3$, the game has a “wary” equilibrium where...*

- *The Receiver plays the “accept-reject” strategy $a_R(2) = 2$, $a_R(3) = 1$.*
- *The aligned-type Sender plays the “partial truth-telling” strategy $r(1) = 1$, $r(2) = r(3) = 2$.*

⁷If the Receiver is recommended to take action 1, their action set is the singleton set $\{1\}$. It is therefore without loss to restrict attention to the Receiver's responses to recommendations of 2 and 3.

⁸Unlike in standard cheap talk (Crawford and Sobel, 1982), the recommendation game does not have a babbling equilibrium: one in which the Sender's messages are completely uninformative, and the Receiver always takes the action that is optimal under their prior. As shown in the proof of Proposition 1, the lack of a babbling equilibrium is due to the assumption that the aligned-type Sender recommends the action closest to $\hat{a}(\theta)$ when indifferent between two or more recommendations.

When $q_3 \geq q_1(1 - p)$, the game has a “trusting” equilibrium where...

- The Receiver plays the “accept all” strategy $a_R(r) = r$ for all $r \in \{2, 3\}$.
- The aligned-type Sender plays the “full truth-telling” strategy $r(\theta) = \theta$ for all $\theta \in \{1, 2, 3\}$.

Furthermore, the wary and trusting equilibria are the unique pure-strategy weak perfect Bayesian equilibria of the recommendation game.

The experiment selects parameters satisfying $q_3 < q_1(1 - p)$; that is, parameters for which only a wary equilibrium exists. I focus on the wary equilibrium because it captures the strategic problem at the heart of this paper. In this equilibrium, the aligned-type Sender must distort their private information not because their preferences conflict with the Receiver’s, but because the Receiver may suspect that they do. Doing so is the only way to induce the Receiver to move away from the status quo, prompting them to take the second-best action instead of the worst action.

Mixed-Strategy Equilibrium. The recommendation game also has one mixed-strategy weak perfect Bayesian equilibrium which only exists for parameters satisfying $q_1 \geq q_3 \geq q_1(1 - p)$. Importantly, it does not exist for the parameters used in the experiment. Proposition 2 in Appendix A.2 characterizes that mixed-strategy equilibrium.

Robustness to Alternative Assumptions. Restricting the Receiver’s action space, assuming equidistant states, and modeling the non-aligned Sender as a non-strategic commitment type help to keep the game simple and easy to understand, especially for aligned-type Senders. This simplicity helps ensure that deviations from partial truth-telling in the experiment reflect failures to account for the Receiver’s wariness, rather than difficulty understanding a complex strategic environment. Appendix A.3 shows that the optimality of the partial truth-telling strategy is not an artifact of these simplifications: it is robust to allowing the Receiver to choose any action (Proposition A.3.1), allowing non-equidistant states (Proposition A.3.2), and modeling the commitment-type Sender as a strategic, self-interested player (Proposition A.3.3). These variations of the recommendation game are discussed in greater detail in Section 5.

2.3 Behavioral Predictions

The experiment selects parameters for which partial truth-telling is the theoretically optimal strategy for participants playing as aligned-type Senders, who I henceforth call “Senders.” I now consider two behavioral mechanisms that may prevent Senders from playing partial truth-telling: incorrect beliefs about the Receiver’s strategy and a failure of contingent thinking.

As shown below, these mechanisms generate different predictions about how Senders should respond to feedback about the Receiver’s strategy and to interventions that make that strategy salient.

The focus on Sender behavior reflects my research objectives and the structure of the recommendation game. As shown in Lemma 1, under the parameters used in the experiment, accept-reject is the Receiver’s unique best response to both full truth-telling and partial truth-telling. Thus, for either pure strategy that Senders could play in a pure-strategy equilibrium, Receivers have the same best response. This makes it plausible that participants playing as Receivers will play accept-reject, so that the main behavioral question is whether Senders best-respond to this Receiver strategy.

LEMMA 1: *Suppose $q_3 < q_1(1 - p)$. Then accept-reject is the Receiver’s unique best response to the belief that the aligned-type Sender is playing either partial truth-telling or full truth-telling.*

Incorrect Beliefs About the Receiver’s Strategy

Senders may fail to play partial truth-telling because they hold incorrect beliefs about the Receiver’s level of strategic sophistication. I formalize this possibility using a level- k model, in which players are assigned strategic types corresponding to their level of strategic reasoning. In the classic level- k model, a level- k player best-responds to a level- $(k - 1)$ player, where $k \geq 1$ (Stahl and Wilson, 1994; Nagel, 1995). In the recommendation game, this framework generates a simple distinction: a level-1 Sender plays full truth-telling, while a level-2 or higher Sender plays partial truth-telling.

To characterize a level-1 Sender’s strategy, one must specify the behavior of a level-0 Receiver. Following a common assumption in level- k analyses of communication games, I assume the level-0 Receiver is credulous: they always take the recommended action (Crawford, 2003; Cai and Wang, 2006; Crawford et al., 2013). A level-1 Sender, who responds to this credulous Receiver, therefore plays full truth-telling. From Lemma 1, it follows that a level-1 Receiver plays accept-reject in response to the level-1 Sender. A level-2 Sender, who responds to a level-1 Receiver, therefore plays partial truth-telling. This level-2 Sender behavior implies that accept-reject is optimal for Receivers of all levels $k \geq 1$, which in turn makes partial truth-telling optimal for Senders of all levels $k \geq 2$.

Thus, if Receivers are level-1 or higher,⁹ a Sender who plays full truth-telling is best-responding to an incorrect belief about the Receiver’s strategy. This mechanism has a clear

⁹Level-0 is often interpreted as an anchor for higher-order beliefs rather than a commonly observed participant type. For example, Stahl and Wilson (1994) estimate an insignificant level-0 share, and Crawford et al. (2013) summarize the literature as suggesting that level-0 participants exist mainly in the minds of higher-level types. When analyzing the experimental results, I directly test the assumption that Receivers are level-1 or higher.

empirical implication in an experiment where Senders receive feedback about the Receiver’s strategy. If a Sender repeatedly receives feedback that the Receiver played accept-reject, and if that Sender initially plays full truth-telling due to incorrect beliefs about the Receiver’s strategy, that Sender should update their beliefs and switch to partial truth-telling.

BEHAVIORAL PREDICTION 1: *Suppose a Sender fails to play partial truth-telling because they incorrectly believe the Receiver will play accept-all. Then feedback indicating that the Receiver plays accept-reject should make the Sender more likely to play partial truth-telling.*

Failure of Contingent Thinking

Alternatively, Senders may fail to play partial truth-telling because of a failure of contingent thinking (FCT). In general, a FCT describes a failure to optimize unless the decision-maker is guided to focus on the contingencies relevant to their decision. [Niederle and Vespa \(2023\)](#) argue that, while contingencies are often hypothetical events (e.g., winning an auction or being pivotal in a committee vote), they may also consist of opponents’ strategies. In strategic settings, such failures can therefore arise from *strategic myopia*: players may fail to account for an opponent’s strategy unless that strategy is made salient. I adapt a simple framework introduced in [Niederle and Vespa \(2023\)](#), which formalizes the concept of a contingency and a failure to think about it, to characterize how a strategic myopia-driven FCT would affect a Sender’s strategy in the recommendation game.

Let $\mathcal{R}$ be the Sender’s pure strategy set, and let $\mathcal{A}$ be the Receiver’s pure strategy set. To capture a failure of contingent thinking driven by strategic myopia, I define a *contingency* as one of the Receiver’s pure strategies $A_i \in \mathcal{A}$. Let $u(R_j, A_i)$ denote the Sender’s utility from playing pure strategy $R_j \in \mathcal{R}$ in contingency A_i . A_i is realized with probability p_i , with $\sum_i p_i = 1$ and $p_i \in [0, 1]$ for each of the Receiver’s four pure strategies $i \in \{1, \dots, 4\}$.

The *relevant contingency* is the contingency with the highest probability of being realized; that is, A_{i^*} such that $p_{i^*} > p_i$ for all $i \in \{1, \dots, 4\} \setminus i^*$. Put differently, it is the pure strategy that the Receiver is most likely to play. I assume for now, and later test in the experimental analysis, that the Receiver’s accept-reject strategy is the unique relevant contingency for Senders in the experiment. A Sender displays a *failure of contingent thinking* if they fail to play the strategy that is optimal for the relevant contingency because that contingency is not salient. Formally, the Sender plays $R_{j'} = \arg \max_{R_j \in \mathcal{R}} u(R_j, A_{i'})$ instead of $R_{j^*} = \arg \max_{R_j \in \mathcal{R}} u(R_j, A_{i^*})$, where $A_{i'} \neq A_{i^*}$ and $R_{j'} \neq R_{j^*}$. In the recommendation game, the strategy that is optimal for the relevant contingency is partial truth-telling.

[Niederle and Vespa \(2023\)](#) discuss behavioral hallmarks of FCTs that follow from this framework. First, decision-makers who display a FCT tend to play a naïve strategy rather than the

optimal one. When the recommendation game’s parameters only admit the wary equilibrium, the natural interpretation of a “naïve” Sender strategy is full truth-telling. This strategy would be optimal if the Receiver were required to follow the Sender’s recommendation; that is, if the Sender did not need to condition on the Receiver’s response. Second, FCTs are more effectively corrected by interventions that place decision-makers in the relevant contingency, or otherwise make that contingency salient, than by feedback about the suboptimal decision. If a FCT drives a Sender’s failure to play partial truth-telling, feedback showing that Receivers play accept-reject should therefore have little effect on the Sender’s likelihood of playing partial truth-telling. This differs from the prediction generated by the incorrect-beliefs mechanism, under which such feedback should induce Senders to switch to partial truth-telling. By contrast, interventions that encourage Senders to focus on the Receiver’s accept-reject strategy should increase partial truth-telling.

BEHAVIORAL PREDICTION 2: Suppose a Sender fails to play partial truth-telling due to a failure of contingent thinking. Then that Sender should (1) play full truth-telling, (2) remain unlikely to play partial truth-telling after receiving feedback that the Receiver plays accept-reject, and (3) become more likely to play partial truth-telling when encouraged to respond to the Receiver’s accept-reject strategy.

3 Main Treatment

The experiment’s primary objective is to determine why Senders may fail to realize that, when a Receiver is wary of their incentives, they should distort their private information to induce the Receiver to trust them. To that end, the experiment implements a setting where Senders must reason in this way: it implements parameters of the recommendation game for which partial truth-telling is the unique equilibrium benchmark.

To make the recommendation game easy for participants to understand, I frame it as a “Plant Pot” Game: Receivers’ goal is to produce the largest possible plant by turning on the sprinkler (i.e., taking an action) that is as close as possible to a pot with a plant seed in it (i.e., the state). They do so by deciding whether to accept a recommendation about which sprinkler to turn on. The recommendation to turn on sprinkler 3 may come either from the computer, which plays the role of the commitment-type Sender, or from another participant, who plays the role of the aligned-type Sender.

The main treatment is designed to distinguish two explanations for failures to account for the Receiver’s wariness. To investigate whether Senders have incorrect beliefs about the Receiver’s strategy, Senders receive feedback about that strategy. To investigate a possible failure

of contingent thinking, Senders play a round in which they directly respond to each Receiver strategy. This final round preserves the strategy spaces and payoffs of the recommendation game while allowing the Sender to directly respond to the Receiver's accept-reject strategy. The design of this round was inspired by an intervention shown to mitigate failures of contingent thinking in committee voting problems (Esponda and Vespa, 2014).

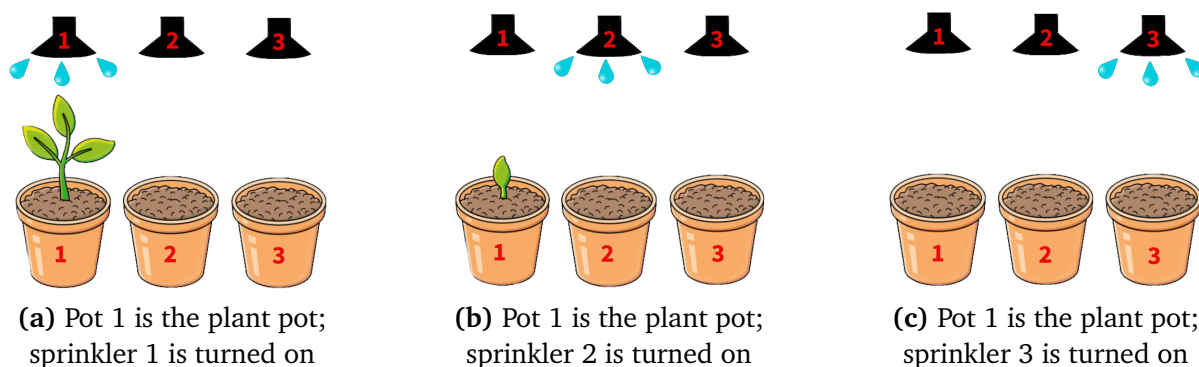
Section 3.1 describes the design of the main treatment. Section 3.2 presents and discusses its results.

3.1 Design

The Plant Pot Game

The Plant Pot Game is the core task in the main treatment. The game features three pots arranged in a line. One pot, the “plant pot,” contains a seed; the other two do not. Each pot has a sprinkler above it. When turned on, a sprinkler waters all pots; however, the closest pot receives the most water, while the farthest pot receives the least. As illustrated in Figure 1, a seed grows into a larger plant when it receives more water.

Figure 1
Illustration of the Plant Pot Game



Receivers are incentivized to produce the largest possible plant: they receive \$16 for producing a large plant, \$13 for producing a small plant, and \$10 for producing no plant. They know that pot 1 has a 9-in-20 chance of being the plant pot, pot 2 has a 7-in-20 chance, and pot 3 has a 4-in-20 chance.

Sprinkler 1 is turned on by default, but Receivers may receive a recommendation to switch to another sprinkler. If recommended to switch, a Receiver chooses whether to accept or reject the recommendation, where rejecting entails turning on the default sprinkler. If recommended to turn on the default sprinkler, the Receiver automatically does so.

Receivers know that they have a 1-in-3 chance of receiving a Sender's recommendation and a 2-in-3 chance of receiving the computer's recommendation. Senders are participants who have the same incentives as Receivers¹⁰ and observe which pot is the plant pot. Senders also know all information available to Receivers when Receivers decide whether to accept a recommendation. The computer recommends switching to sprinkler 3 regardless of which pot is the plant pot.

The Plant Pot Game implements the recommendation game with a linear payoff function for the Receiver and Sender, and parameters for which only the wary equilibrium exists. Thus, it is optimal for Receivers to play their accept-reject strategy: accept a recommendation to switch to sprinkler 2 and reject a recommendation to switch to sprinkler 3. Senders should therefore respond by playing their partial truth-telling strategy: recommend the sprinkler corresponding to the plant pot when the plant pot is pot 1 or 2, but recommend sprinkler 2 when the plant pot is pot 3.

Treatment Structure

Experimental sessions were conducted with even numbers of participants. At the beginning of a session, half of the participants were randomly assigned to each role (Sender or Receiver). Participants retained their roles throughout the experiment.

Figure 2
Structure of the Main Treatment

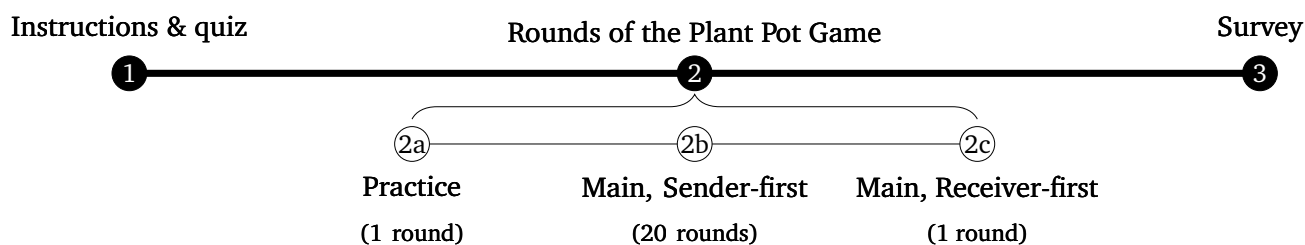


Figure 2 summarizes the structure of the main treatment. After reading the instructions, participants completed a comprehension quiz. They were required to answer all quiz questions correctly before proceeding to the first round. To incentivize careful reading, participants received a \$1 bonus if they answered all questions correctly on their first attempt.

Participants then played 22 rounds of the Plant Pot Game. The first round was an unpaid practice round; the remaining rounds, which I call the “main” rounds, were eligible for payment. At the beginning of each round, Senders and Receivers were randomly re-matched,

¹⁰If a Receiver received the computer's recommendation, the Sender they were paired with received \$10 regardless of the size of the plant the Receiver produced.

and the plant pot and source of the Receiver’s recommendation were drawn from the relevant probability distributions.

Participants’ strategies were elicited using the strategy method. In every round, Receivers specified whether they would accept or reject each recommendation involving a switch from the default sprinkler. This allowed me to provide Senders with feedback about how the Receiver they were paired with would have responded to each recommendation involving a switch, regardless of the recommendation the Receiver actually received.¹¹ Players in both roles received end-of-round feedback about their co-player’s strategy and the outcome of the round.

Sender strategies were elicited differently across rounds. In the practice round and first 20 main rounds, Senders specified the sprinkler they would recommend for each possible plant pot. These rounds followed the sequence of the recommendation game in Section 2: Senders made recommendations, then Receivers responded. In the final main round, the sequence was reversed: Receivers first committed to a response to each possible recommendation, then Senders specified the sprinkler they would recommend for each possible plant pot and each possible Receiver strategy: accept all, accept-reject, reject-accept, and reject all. I refer to the first 20 main rounds as Sender-first rounds and the final main round as the Receiver-first round.

After completing all rounds of the Plant Pot Game, participants completed an end-of-experiment survey, which included questions about the rationale behind their strategies and a demographic questionnaire.

Procedures

Appendix B contains the main treatment’s instructions. The treatment was pre-registered on AsPredicted.org, programmed and deployed in oTree ([Chen et al., 2016](#)), and conducted at the Toronto Experimental Economics Laboratory (TEEL). Participants were recruited from the University of Toronto student body using the ORSEE online recruitment system ([Greiner, 2015](#)). I conducted seven sessions, each with 16 to 20 participants, for a total of 136 participants (68 Senders, 68 Receivers).

Participants’ total payment included a \$5 show-up fee, their earnings from the comprehension quiz, and their earnings from one main round. Each participant’s paid round was randomly selected at the beginning of the experiment and revealed at the end. The average participant earned \$18.69 and took 40 minutes to complete the experiment.

¹¹I use the strategy method to elicit participants’ strategies because, unlike the direct response method, it allows me to give Senders feedback about Receivers’ full strategies. This feedback is important for testing whether communication errors arise from incorrect beliefs: it ensures that, at the end of each round, all Senders have the opportunity to learn how a Receiver would have responded to a recommendation to switch to sprinkler 3. Studies that have compared the strategy and direct response methods have shown that participants’ responses do not tend to differ based on the method used ([Brandts and Charness, 2011](#)).

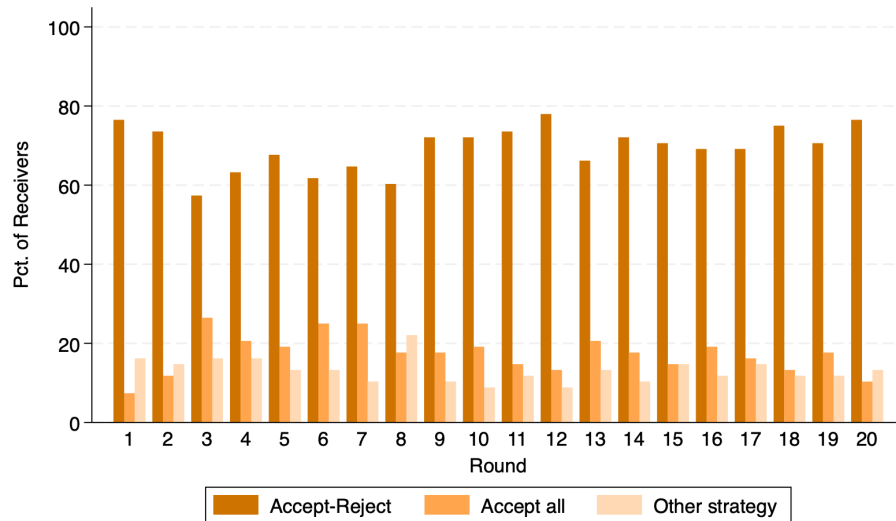
3.2 Results

I begin by analyzing Receivers' strategies, to confirm that partial truth-telling is empirically optimal for Senders. Next, I investigate why Senders may fail to play that strategy. Unless otherwise specified, all analyses focus exclusively on the 20 Sender-first rounds. I exclude one participant who withdrew from the experiment,¹² which yields a final sample of 135 participants (67 Senders, 68 Receivers).

Receivers' Strategies

Given the parameters used in the experiment, it is theoretically optimal for Senders to play partial truth-telling; that is, to recommend the action corresponding to the state when the state is 1 or 2, and to recommend 2 when the state is 3. I first study whether this strategy is also empirically optimal. For this to be the case, a majority of Receivers must play accept-reject: accept a recommendation of 2 and reject a recommendation of 3. I therefore begin by analyzing Receivers' strategies and the feedback Senders receive about them.

Figure 3
Distribution of Receivers' Strategies



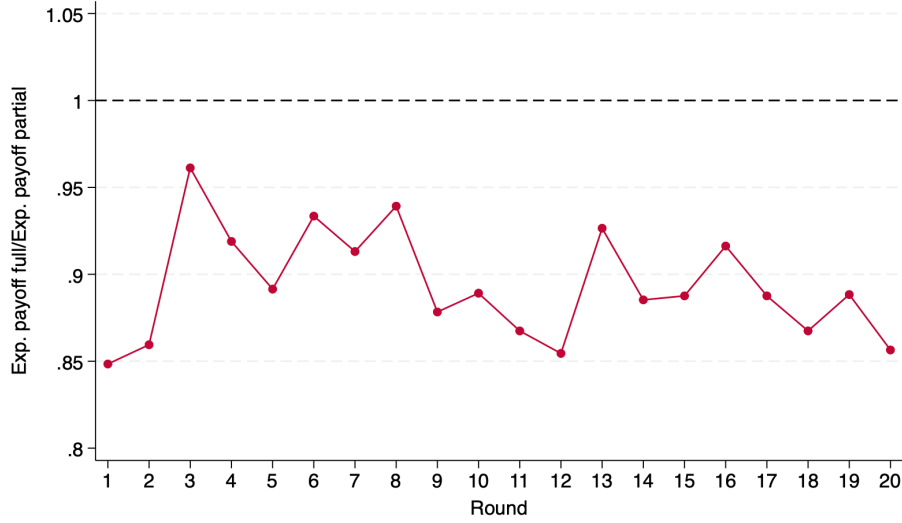
Notes. This figure displays the percent of Receivers in the main treatment playing accept-reject (their wary equilibrium strategy), accept-all (their trusting equilibrium strategy), and other strategies in a given round.

Consistent with the theoretical prediction, a clear majority of Receivers plays accept-reject. Figure 3 displays the distribution of Receivers' strategies in each round. In the average round,

¹²This participant was playing the role of Sender, and withdrew during the survey.

70% of Receivers play accept-reject, while at most 27% of Receivers play any other strategy in any given round. As a result, accept-reject is the modal Receiver strategy in every round.¹³

Figure 4
Expected State-3 Payoff of Senders' Equilibrium Strategies



Notes. This figure displays the expected state-3 payoff of Senders' full truth-telling strategy as a fraction of the expected state-3 payoff of their partial truth-telling strategy. Expected payoffs are calculated using the empirical distribution of Receivers' strategies in the main treatment.

Partial truth-telling is indeed the best response to this empirical distribution of Receivers' strategies. Since Senders' equilibrium strategies differ only in the action recommended in state 3, Figure 4 compares the expected state-3 payoff of these strategies,¹⁴ given the empirical distribution of Receivers' strategies. In all rounds, partial truth-telling yields a higher expected payoff than full truth-telling: the index $\frac{\text{Exp. payoff of full truth-telling}}{\text{Exp. payoff of partial truth-telling}}$ is always below 1. Across all rounds, the expected payoff of partial truth-telling is up to 15% higher, with an average payoff difference of 11%.¹⁵

¹³Since the experiment uses values of q_1 , q_3 and p such that $q_3 < q_1(1 - p)$, accept-reject is the unique optimal strategy for Receivers (see Lemma 1). Thus, even if most Senders play full truth-telling instead of partial truth-telling, accept-reject is empirically optimal for Receivers.

¹⁴Let $P_{a_R(2)a_R(3)}$ be the percent of Receivers who play strategy $(a_R(2), a_R(3))$. The expected state-3 payoff to full truth-telling is $(P_{21} + P_{11}) \cdot 10 + (P_{23} + P_{13}) \cdot 16$. The expected state-3 payoff to partial truth-telling is $(P_{21} + P_{23}) \cdot 13 + (P_{11} + P_{13}) \cdot 10$.

¹⁵The maximum and average percent payoff differences correspond to \$1.90 and \$1.41 absolute payoff differences, respectively. To put these absolute differences into perspective, recall that participants received a \$1 bonus payment for answering all comprehension questions correctly on their first attempt. This suggests that payoff differences that were up to 90% larger than the bonus payment would have been meaningful to participants.

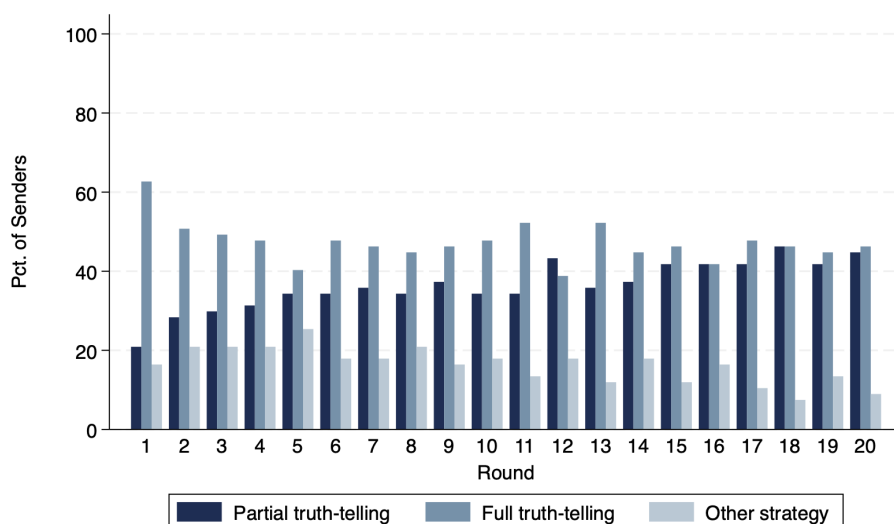
RESULT 1: *Partial truth-telling is Senders' empirically-optimal strategy.*

Furthermore, the fact that Receivers consistently play accept-reject is reflected in the feedback Senders receive. Appendix Figure C.1 displays, for rounds 2 through 20, the percent of previous rounds in which the average Sender received the feedback that they were paired with a Receiver who played accept-reject. The average Sender has received this feedback in 66% to 76% of prior rounds. The average frequency across all rounds is 69% (roughly 14 out of the 20 rounds), which is essentially identical to the true average frequency with which Receivers play accept-reject.

To summarize, partial truth-telling is empirically optimal, and Senders have the opportunity to learn this. Next, I investigate whether Senders play this empirically-optimal strategy.

Senders' Strategies

Figure 5
Distribution of Senders' Strategies



Notes. This figure displays the percent of Senders in the main treatment playing partial truth-telling (their wary equilibrium strategy), full truth-telling (their trusting equilibrium strategy), and other strategies in a given round.

Figure 5 displays the distribution of Senders' strategies in each round. Focusing first on the frequency of partial truth-telling, in any given round, less than half of Senders play this strategy. The proportion doing so begins at 21% in the first round, reaches a maximum of 46% in round 18, and settles at 45% in the final round. Put differently, only a minority of Senders

play their empirically-optimal strategy in any given round.

A similar result emerges when analyzing Senders' strategies at the player level. Appendix Figure C.2a displays the proportion of Senders playing partial truth-telling in a given number of the 20 rounds. The median frequency is two rounds. Perhaps more strikingly, the modal frequency - which accounts for 37% of Senders' behavior - is zero rounds. Appendix Figures C.2b and C.2c repeat the analysis from Figure C.2a, but focus on Senders' behavior in the last half and quarter of the 20 rounds, respectively. The figures show that, even with extensive experience, the modal Sender plays partial truth-telling in zero rounds.

Instead of playing the empirically-optimal partial truth-telling strategy, many Senders play full truth-telling; that is, the strategy that would be optimal if Receivers accepted all recommendations. Figure 5 shows that full truth-telling is almost always the most popular Sender strategy. It is the modal Sender strategy in 19 out of the 20 rounds, and 40% to 63% of Senders play it in any given round.

RESULT 2: A minority of Senders play partial truth-telling in any given round. Full truth-telling is the modal Sender strategy in almost all rounds.

Having established that Senders often fail to play their empirically-optimal strategy, I now investigate the root of this failure. One possible explanation is that Senders incorrectly believe that Receivers will accept all recommendations. As summarized in Behavioral Prediction 1, if this were the case, then receiving feedback that Receivers play accept-reject should help correct these beliefs and induce Senders to play partial truth-telling. Since Senders accumulate such feedback over the course of the experiment, the incorrect-beliefs mechanism requires Senders to become meaningfully more likely to play partial truth-telling in later rounds, and especially after receiving a larger amount of feedback that Receivers play accept-reject.

Table 1
Evolution of Senders' Likelihood of Playing Partial Truth-Telling (Main Treatment)

<i>Rounds</i>	All		2-10		11-20	
	(1)	(2)	(3)	(4)	(5)	(6)
Round	0.008*** (0.003)	0.008*** (0.003)	0.009 (0.007)	0.010 (0.006)	0.009** (0.004)	0.009** (0.004)
Prev.-round feedback		0.005 (0.077)		0.054 (0.077)		0.261 (0.198)
Constant	0.594*** (0.031)	0.589*** (0.059)	0.280*** (0.040)	0.231*** (0.065)	0.860*** (0.065)	0.652*** (0.191)
Observations	1273	1273	603	603	670	670
Mean of DV	0.37	0.37	0.33	0.33	0.41	0.41

Notes. Coefficients from OLS regressions conducted among Senders in the main treatment, with standard errors in parentheses. The dependent variable is an indicator for playing partial truth-telling in a given round. In Columns (1), (3) and (5), round number is the only independent variable. In Columns (2), (4) and (6), independent variables include the round number and the fraction of previous rounds in which the Sender was paired with a Receiver who played accept-reject. All regressions include Sender-fixed effects and standard errors clustered at the Sender-level. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$.

Table 1 reports regressions that study Senders' likelihood of playing partial truth-telling as a function of the round number and prior feedback that Receivers play accept-reject. Across all 20 rounds, the coefficient on Round is positive and statistically significant, but small: each additional round is associated with a 0.8 percentage point increase in the probability that a Sender plays partial truth-telling. This estimate is essentially unchanged when prior feedback is included, suggesting that the gradual increase in partial truth-telling over time is not meaningfully explained by feedback about Receivers' use of accept-reject. When the sample is split into early and later rounds, the round coefficient is only statistically distinguishable from zero in later rounds, although the estimated magnitudes are similar across the two subsamples. More importantly, the feedback coefficients are positive but imprecisely estimated and statistically insignificant in every specification: Senders do not respond strongly to information that Receivers are playing accept-reject. This suggests that incorrect beliefs are unlikely to be the root cause of Senders' failure to play partial truth-telling.

RESULT 3: *Senders become only slightly more likely to play partial truth-telling as rounds progress, and feedback that Receivers play accept-reject does not significantly predict Sender behavior.*

I next consider whether Senders' communication failure instead reflects a strategic myopia-

driven failure of contingent thinking (FCT). As a reminder, this captures a failure to best-respond to the strategy the Receiver is most likely to play (accept-reject) unless that strategy is made salient.

In fact, the Sender behavior documented so far is consistent with two of the three behavioral hallmarks of FCTs outlined in Behavioral Prediction 2. First, as summarized by Result 2, many Senders play the “naïve” full truth-telling strategy instead of the optimal partial truth-telling strategy. Second, as summarized by Result 3, Senders do not react strongly to feedback that Receivers are playing accept-reject.

The third hallmark is that decision-makers can often overcome an FCT when placed in the relevant contingency or encouraged to focus on it. The Receiver-first round tests the effectiveness of such an intervention. In this round, Senders specify a recommendation for each possible Receiver strategy. In particular, they directly respond to the Receiver’s accept-reject strategy, focusing their attention on the Receiver strategy that arises in the wary equilibrium. If Senders’ communication failure is driven by a FCT, Senders should therefore be substantially more likely to play partial truth-telling when responding to accept-reject in the Receiver-first round.

Table 2
Senders’ Strategies: Final Sender-First Round vs. Receiver-First Round

	<u>Response to accept-reject</u>		
	Partial truth-telling	Other strategy	<i>Total</i>
<u>Final Sender-first round</u>			
Partial truth-telling	86.7%	13.3%	100.0%
Other strategy	70.3%	29.7%	100.0%
<i>Total</i>	<i>77.6%</i>	<i>22.4%</i>	<i>100.0%</i>

Notes. This table displays main treatment Senders’ strategies in the Receiver-first round, as a function of the strategies they play in the final Sender-first round. The cells display the proportion of Senders who play a given strategy in the final Sender-first round, who then play a strategy shown in a given column in response to accept-reject in the Receiver-first round.

Table 2 compares Senders’ strategies in the final Sender-first round to their responses to accept-reject in the Receiver-first round. Recall that the Receiver-first round occurs immediately after the final Sender-first round, so Senders do not gain additional experience with the game between the two rounds. Nevertheless, Senders are significantly more likely to play partial truth-telling when directly responding to accept-reject. As shown in the final row of Table 2, nearly 78% of Senders play partial truth-telling in the Receiver-first round, compared with 45% in the final Sender-first round (proportions test, $p < 0.01$). This increase is driven by

Senders who do not play partial truth-telling in the final Sender-first round: over 70% of these Senders switch to partial truth-telling when responding to accept-reject. In contrast, almost all Senders who play partial truth-telling in the final Sender-first round continue to do so when responding to accept-reject.

RESULT 4: Senders are much more likely to play partial truth-telling when directly responding to the Receiver's accept-reject strategy.

The evidence accumulated thus far suggests that Senders' communication failure is better explained by a failure of contingent thinking than by incorrect beliefs about Receivers' behavior. I next show that the failure is broadly inconsistent with two additional mechanisms that could also lead Senders to play full truth-telling too often.

1. Strict truth-telling preferences. People sometimes exhibit a strict preference for truth-telling (Abeler et al., 2019). If such a preference explained Senders' behavior, then Senders should be reluctant to play any strategy that requires recommending an action that does not correspond to the state. In particular, they should play full truth-telling regardless of the Receiver's response. The data are inconsistent with this behavioral pattern: only 6% of Senders play full truth-telling in response to every Receiver strategy in the Receiver-first round. Thus, while a strict preference for truth-telling may explain the behavior of a small minority of Senders, it cannot account for most Senders' communication failure.

2. Mixing between partial and full truth-telling. A second possibility is that Senders are mixing between full and partial truth-telling. As discussed in Section 2.2, the recommendation game can have a mixed-strategy equilibrium, but this equilibrium does not exist under the parameters used in the experiment. Moreover, the data provide very limited evidence that Senders are mixing. Appendix Table C.1 classifies Senders by strategy type, where a type is defined by the frequency with which Senders use different strategies across the 20 Sender-first rounds. Regardless of the strictness of the classification rule, very few Senders (at most 9%) are classified as mixing between partial and full truth-telling. Instead, roughly three-quarters of Senders consistently play either full truth-telling or partial truth-telling. Thus, Senders do not appear to play full truth-telling more often than predicted because they are mixing between full and partial truth-telling.

Main Treatment Results: Summary and Implications

The main treatment yields three key findings. First, although partial truth-telling is empirically optimal, only a minority of Senders play it in any given round. Instead, Senders tend to play full truth-telling: the strategy that would be optimal if Receivers accepted all recommendations. Second, feedback about Receivers' strategies has little effect, and most Senders' behavior is inconsistent with strict truth-telling preferences or mixing between partial and full truth-telling. Third, Senders are much more likely to play partial truth-telling when they directly respond to the Receiver's accept-reject strategy. Together, these findings are most consistent with a failure of contingent thinking: Senders often play a naïve strategy and respond weakly to feedback, but respond strongly when the relevant Receiver strategy is made explicit.

While the main treatment's results strongly suggest that Senders' communication failure stems from a failure of contingent thinking, I conduct a second treatment to obtain additional data that separates belief and reasoning errors. As explained below, the second treatment accomplishes two goals. First, it yields data on Senders' strategic beliefs, providing a further test of the incorrect-beliefs mechanism. Second, it implements an intervention that should make the Receiver's accept-reject strategy more salient, providing a further test of the failure of contingent thinking mechanism.

4 Beliefs Treatment

In the beliefs treatment, Senders report their beliefs about the modal Receiver's strategy before choosing their own strategy. The treatment is designed both to measure beliefs and to make those beliefs decision-relevant at the moment of choice. If Senders fail to play partial truth-telling because they do not know how Receivers behave, the belief data should reveal widespread incorrect beliefs. If instead Senders fail because they neglect to account for the Receiver's response, then eliciting beliefs should increase partial truth-telling by making the relevant Receiver strategy salient, as indicated by the third behavioral hallmark from Behavioral Prediction 2.

Section 4.1 describes the design of the beliefs treatment. Section 4.2 presents and discusses its results.

4.1 Design

Treatment Structure

The beliefs treatment preserves the structure of the main treatment as closely as possible, while adding belief elicitation before selected rounds. As in the main treatment, participants played 22 rounds of the Plant Pot Game, with the first round serving as an unpaid practice round and the remaining rounds eligible for payment. The only difference was that, before certain Sender-first rounds, participants reported beliefs about their co-player’s strategy in the upcoming round.

More precisely, participants reported their belief about the modal strategy played by participants in the other role in previous sessions of the experiment. Senders reported their beliefs about Receivers’ modal strategy. Receivers reported their beliefs about Senders’ modal strategy and also re-stated the recommendation the computer always sent. I elicited beliefs from participants in both roles to preserve procedural symmetry across Senders and Receivers, and to keep the length and structure of the experiment similar across roles. This design choice also made it possible to assess whether the belief-elicitation task affected behavior differently across roles.

Beliefs were elicited before every sixth Sender-first round, starting with the first one. Thus, there were four belief-elicitation tasks, which occurred before the first, seventh, thirteenth, and nineteenth Sender-first rounds. Beliefs were elicited in only a subset of rounds to keep the beliefs treatment comparable in length to the main treatment and to avoid making belief elicitation the dominant feature of the experiment.

The belief elicitation tasks were incentivized using a simple scheme that paid \$5 if the stated belief was correct and nothing otherwise.¹⁶ I used this simple incentive scheme rather than a scoring rule because complex elicitation schemes may confuse participants (Danz et al., 2022), and because a key goal of the beliefs treatment was to make the key Receiver-strategy contingency salient. Eliciting full probability distributions over Receiver strategies may have distracted from that contingency.

Procedures

Appendix B contains the beliefs treatment’s instructions. As in the main treatment, the beliefs treatment was pre-registered on AsPredicted.org, programmed and deployed in oTree

¹⁶For payment purposes, the correct modal strategies were the modal strategies in the main treatment. As a reminder, in all rounds of the main treatment, the modal Receiver played accept-reject, and the modal Sender played full truth-telling. In addition to correctly stating the Sender’s strategy, Receivers were also required to correctly re-state the computer’s recommendation in order to receive the \$5 payment.

(Chen et al., 2016), and conducted at TEEL with participants recruited via ORSEE (Greiner, 2015). I conducted seven sessions, each with 16 to 22 participants, for a total of 130 participants (65 Senders, 65 Receivers).

Participants' total payment included a \$5 show-up fee, their earnings from the comprehension quiz, their earnings from one main round, and their earnings from one belief-elicitation task. Each participant's paid round and belief-elicitation task were randomly selected at the beginning of the experiment and revealed at the end. The average participant earned \$21.71 and took 41 minutes to complete the experiment. Appendix Table C.2 shows that participants' characteristics did not differ significantly between the main and beliefs treatments.

4.2 Results

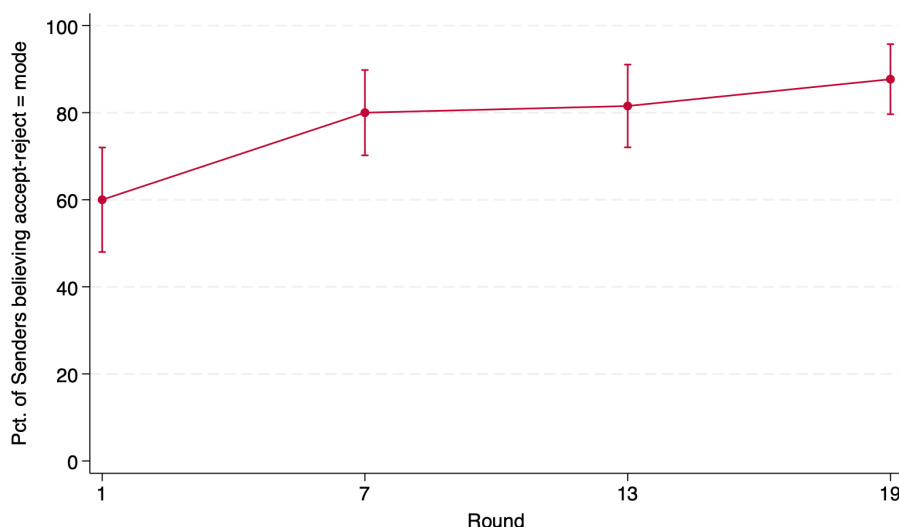
Given the objectives of the beliefs treatment, I focus primarily on Senders' beliefs and strategies. I first analyze Senders' beliefs about the modal Receiver strategy. This analysis establishes whether most Senders anticipate that Receivers tend to play accept-reject, and thus whether belief elicitation makes the relevant Receiver-strategy contingency salient. I then compare strategies across treatments. If Senders' communication failure stems from a failure of contingent thinking, then Senders who correctly believe that the modal Receiver plays accept-reject should be more likely to play partial truth-telling after reporting that belief.

As in the main treatment, I focus on the 20 Sender-first rounds and drop participants who withdrew from the experiment.¹⁷ The final sample includes 129 participants (65 Senders, 64 Receivers). All reported p -values are from two-sided tests.

¹⁷One participant withdrew from the beliefs treatment. They were playing the role of Receiver and withdrew after round 11. Because one-to-one matching of Receivers to Senders requires an even number of participants to be present in all rounds, a research assistant played at this participant's terminal for the remainder of the session. The research assistant played the modal Receiver strategy from all prior experimental sessions, which was accept-reject.

Senders' Beliefs

Figure 6
Senders' Beliefs about Receivers' Strategies



Notes. This figure displays the percent of Senders in the beliefs treatment who believe that, in prior experimental sessions, the modal Receiver strategy in round x was accept-reject. Bars indicate 95% confidence intervals.

Figure 6 displays the share of Senders who believe that accept-reject is the modal Receiver strategy in each belief-elicitation round. In every such round, most Senders hold this correct belief. Moreover, this share increases across rounds: between rounds 1 and 19, the share of Senders with correct beliefs increases from 60% to 88%. This increase, as well as the increase from round 1 to round 7, is statistically significant (proportions tests, $p \leq 0.01$ for both).¹⁸

RESULT 5: *In every belief-elicitation round, most Senders correctly believe that accept-reject is the modal Receiver strategy.*

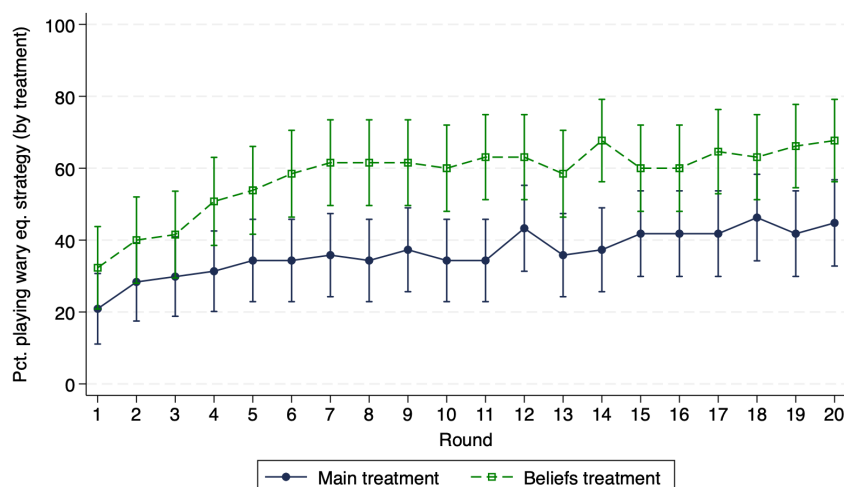
Result 5 has two implications for interpreting Sender behavior. First, the fact that most Senders correctly identify the modal Receiver strategy further weakens the possibility that in-

¹⁸Appendix Table C.3 documents how Receivers' beliefs about Senders' strategies evolve across rounds. Changes in Receivers' beliefs track empirical trends in Senders' strategies: in round 1, the modal Receiver believes full truth-telling is the most common Sender strategy; by round 19, this modal belief has shifted to partial truth-telling. As shown in Lemma 1, for the parameters used in the experiment, accept-reject is a Receiver's unique best response to the belief that an aligned-type Sender is playing either full truth-telling or partial truth-telling. Appendix Figure C.3, which shows the share of Receivers playing accept-reject in the beliefs treatment, therefore indicates that the vast majority of Receivers best-respond to their beliefs.

correct strategic beliefs are the primary cause of failures to play partial truth-telling. Second, because beliefs are elicited immediately before Senders choose their strategy, the elicitation makes the relevant contingency - the Receiver's accept-reject strategy - salient at the moment of choice. Next, I study whether this intervention increases partial truth-telling.

Senders' Strategies

Figure 7
Senders' Frequency of Optimal Strategy Play (Round-Level), by Treatment



Notes. This figure displays the percent of Senders in each treatment playing partial truth-telling (their wary equilibrium strategy) in each round. Bars indicate 95% confidence intervals.

Figure 7 compares the share of Senders playing partial truth-telling in each round, by treatment. In both treatments, the share of Senders playing partial truth-telling increases across rounds. However, this share is consistently higher in the beliefs treatment, and the difference is statistically significant from round 4 onward (proportions tests, $p \leq 0.05$). This pattern is consistent with belief elicitation helping Senders best-respond to Receivers: the treatment effect is larger in later rounds, as more Senders report the correct belief about the modal Receiver strategy.

The treatment effect is robust to using a stricter definition of strategic behavior. In Appendix Figure C.4, following an approach used by [Esponda and Vespa \(2014\)](#), I define a Sender as “strategic” in round x if they play partial truth-telling in round x and all subsequent rounds. As shown in the figure, the distribution of strategic Senders in the beliefs treatment first-order stochastically dominates the distribution of strategic Senders in the main treatment (Wilcoxon rank-sum test, $p < 0.01$).

Receivers' strategies provide a useful benchmark for interpreting this treatment effect. Recall that Receivers' beliefs about Senders' strategies were also elicited every sixth round. Appendix Figure C.3 shows that, unlike for Senders, belief elicitation had little effect on Receivers' likelihood of playing their wary equilibrium strategy. In almost all rounds, the share of Receivers playing accept-reject does not differ significantly across treatments.¹⁹ This finding helps rule out two alternative interpretations of the Sender treatment effect. First, because Receiver behavior changes little across treatments, it is unlikely that Senders in the beliefs treatment received systematically different feedback about Receivers' strategies. Second, the Sender treatment effect is unlikely to be a purely mechanical consequence of reporting beliefs.

The treatment comparison shows that belief elicitation increases partial truth-telling. I next study whether this increase is driven by Senders who report that accept-reject is the modal Receiver strategy. If belief elicitation focuses Senders on the relevant contingency, partial truth-telling should be most common among Senders with correct beliefs.

Table 3
Effect of Beliefs on Senders' Likelihood of Playing Partial Truth-Telling

<i>Rounds</i>	All (1)	Rounds 1 & 7 (2)	Rounds 13 & 19 (3)
Believes accept-reject is mode	0.446*** (0.075)	0.377*** (0.096)	0.541*** (0.096)
Constant	0.017 (0.881)	0.017 (0.897)	-0.068 (0.666)
Observations	260	130	130
Mean of DV	0.546	0.469	0.623

Notes. Coefficients from OLS regressions conducted among Senders in the beliefs treatment, with standard errors in parentheses. The dependent variable is an indicator for playing partial truth-telling (Senders' wary equilibrium strategy) in a given round, and the independent variable is an indicator for believing accept-reject (Receivers' wary equilibrium strategy) is the modal Receiver strategy in a given round. All regressions include controls for the round and session. Standard errors clustered at the participant-level. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$.

Pooling the rounds in which beliefs are elicited, correctly predicting the modal Receiver strategy is positively correlated with playing partial truth-telling ($\rho = 0.43$, $p < 0.01$). Table 3 shows that this relationship persists after controlling for the round. As shown in column 1, Senders who report that accept-reject is the modal Receiver strategy in the upcoming round are 44.6 percentage points more likely to play partial truth-telling in that round. Columns 2

¹⁹The share of Receivers playing accept-reject in the beliefs treatment differs from the main treatment in only two of the 20 rounds: rounds 4 ($p < 0.01$) and 8 ($p < 0.05$).

and 3 show that this relationship is present in both early and later rounds. Because most Senders report the correct modal Receiver strategy, this pattern is not simply evidence that Senders lack the information needed to play partial truth-telling. Instead, consistent with a failure of contingent thinking, it suggests that making the relevant Receiver strategy explicit helps Senders identify their optimal strategy.

RESULT 6: Senders are more likely to play partial truth-telling when they focus on responding to Receivers' accept-reject strategy.

Beliefs Treatment Results: Summary and Implications

The beliefs treatment's results provide further evidence that Senders' communication failure reflects a failure of contingent thinking. If the failure were driven primarily by incorrect beliefs about the Receiver's strategy, we would expect most Senders to hold incorrect beliefs about the modal Receiver strategy. If it were driven by strict truth-telling preferences or by mixing between partial and full truth-telling, then the belief elicitation task should not significantly impact Senders' strategies. Instead, most Senders correctly predict that accept-reject is the modal Receiver strategy (Result 5), and focusing on this Receiver strategy makes them significantly more likely to play partial truth-telling (Result 6). Thus, across both treatments, directly or indirectly placing Senders in the relevant contingency reduces their likelihood of playing a naïve strategy (full truth-telling), and increases their likelihood of playing the optimal strategy (partial truth-telling). This pattern is consistent with a failure of contingent thinking.

5 Discussion and Conclusion

Summary and Implications. When Senders communicate, they often have asymmetric information about both the state of the world and their preferences. This paper provides experimental evidence on why Senders may fail to communicate persuasively in such a setting. I propose a novel communication game where the Receiver believes the Sender's preferences may be misaligned with their own, while the Sender knows that their preferences are in fact aligned. I find that, while Senders should distort their private information to improve outcomes for both the Receiver and themselves, many Senders do not do this in practice. Using an experiment that distinguishes several possible behavioral mechanisms, I show that the communication failure is driven by a reasoning error: Senders initially fail to consider how a Receiver will respond to their recommendations, but once prompted to do so, they are able to communicate effectively. While earlier work points to incorrect strategic beliefs as a leading cause

of over-communication, these findings imply that this mechanism may not universally explain all communication failures. More broadly, however, the results point to a strategic reasoning error that can arise even when players’ incentives are aligned and their beliefs about others’ strategies are correct.

Robustness of Optimal Communication Benchmark. In reality, the settings in which recommendations are issued may be more complex than the setting studied here. In Appendix A.3, I show that the optimality of the aligned-type Sender’s partial truth-telling strategy is robust to weakening several assumptions made in the primitives of the recommendation game. These include relaxing the restriction on the Receiver’s action set (Proposition A.3.1), allowing states to be non-equidistant (Proposition A.3.2), and letting the non-aligned Sender be a strategic, self-interested player (Proposition A.3.3).

However, the optimality of a partial truth-telling strategy does require two restrictions on the Receiver’s beliefs and action set. First, the Receiver must have a second-best action that is between their status quo and the first-best action they are wary of taking (see Proposition A.3.4 in Appendix A.3). Put differently, the improvement in the Receiver’s wellbeing relative to their status quo must be monotonically increasing in the size of their behavioral change. This is true in many natural settings, such as those involving health-related outcomes. For instance, while the Canadian government recommends drinking no more than two alcoholic beverages per week, they also state that “any reduction in alcohol use has benefits” (Paradis et al., 2023). Thus, while heavy drinkers would benefit the most from reducing their consumption to the two-drink threshold, a smaller reduction would also be beneficial. Similarly, the World Health Organization’s recommendations for weekly minutes of physical activity are coupled with the statement that “[d]oing some physical activity is better than doing none” (World Health Organization, 2020).

The optimality of a partial truth-telling strategy also requires the Receiver to be skeptical about their need to make a large behavioral change ($q_1 \geq q_3$; see Proposition 1). This assumption is consistent with two commonly-observed behavioral biases: status quo bias (Samuelson and Zeckhauser, 1988) and a form of motivated reasoning that involves underweighting evidence against the need to change one’s behavior (Epley and Gilovich, 2016).

Future Work. My findings point to several interesting avenues for future work. For one, they indicate that simply prompting decision-makers to think about a relevant contingency may help correct a failure of contingent thinking. This intervention is weaker than those studied in prior work, which involve re-framing the decision problem (Esponda and Vespa, 2014; Niederle and Vespa, 2023; Esponda and Vespa, 2024). Studying the effectiveness of this intervention in

other strategic settings may be a promising avenue for future work. More broadly, though, my findings raise questions about the practice of eliciting players' beliefs about other players' strategies. Obtaining data on these beliefs can provide valuable insight into a player's level of strategic sophistication (see, e.g., [Aoyagi et al. \(2024\)](#)). My results suggest, however, that eliciting those beliefs may change a player's strategy. Future research may study the impact of belief elicitation in a broader class of games, to better understand when this practice affects players' strategies.

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Appendix for “Strategic Communication under Perceived Misalignment of Interests”

A Proofs

In all proofs, I use $t \in \{A, C\}$ to denote the Sender’s type, where A indicates the aligned type and C indicates the commitment type. I use $\mu(\theta, t|r)$ to denote the Receiver’s belief about the probability of the history (θ, t) having occurred, conditional on receiving recommendation r .

A.1 Proposition 1 and Lemma 1

Note that if the Receiver is recommended to take action 1 (i.e., to stick with their status quo), their action set is the singleton set $\{1\}$. It is therefore without loss to restrict attention to the Receiver’s responses to recommendations of 2 and 3. Furthermore, given that the Sender’s commitment type is non-strategic, the Receiver and the Sender’s aligned type are the only players in the recommendation game. When describing each equilibrium, I therefore only include the Receiver and aligned-type Sender’s strategies.

A.1.1 Proof of Proposition 1

I first show that the trusting and wary equilibria exist under the proposed parameter conditions, then show that no other pure-strategy weak perfect Bayesian equilibria exist.

1. *Proof that the trusting equilibrium exists.* Suppose the Receiver plays $a_R(r) = r$ for all $r \in \{2, 3\}$: they accept all recommendations to take an action other than their status quo. The aligned-type Sender’s unique best response is $r(\theta) = \hat{a}(\theta)$ for all θ , since they want the Receiver’s action to be as close as possible to $\hat{a}(\theta)$. If the aligned-type Sender uses this strategy, the Receiver forms the following beliefs:

$$\mu(1, C|3) = \frac{q_1(1-p)}{q_1(1-p) + q_2(1-p) + q_3} \quad (1)$$

$$\mu(2, C|3) = \frac{q_2(1-p)}{q_1(1-p) + q_2(1-p) + q_3} \quad (2)$$

$$\mu(3, C|3) = \frac{q_3(1-p)}{q_1(1-p) + q_2(1-p) + q_3} \quad (3)$$

$$\mu(3, A|3) = \frac{q_3 p}{q_1(1-p) + q_2(1-p) + q_3} \quad (4)$$

$$\mu(\theta, A|r) = 1 \text{ for } (\theta, r) \in \{(1, 1), (2, 2)\} \quad (5)$$

$$\mu(\theta, A|r) = 0 \text{ for all } r \neq \theta \quad (6)$$

Given these beliefs, it is always better for the Receiver to accept a recommendation of 2 than to stick with their status quo. Accepting a recommendation of 3 is better than sticking with their status quo if

$$\frac{1}{q_1(1-p) + q_2(1-p) + q_3} [q_1(1-p)u(2\alpha) + q_2(1-p)u(\alpha) + q_3u(0)] \geq \frac{1}{q_1(1-p) + q_2(1-p) + q_3} [q_1(1-p)u(0) + q_2(1-p)u(\alpha) + q_3u(2\alpha)]$$

which simplifies to

$$q_3 \geq q_1(1-p) \quad (7)$$

Thus, if Equation 7 holds, there exists a “trusting” pure-strategy weak perfect Bayesian equilibrium where the aligned-type Sender plays $r(\theta) = \hat{a}(\theta)$ for all θ , the Receiver plays $a_R(r) = r$ for all r , and the Receiver holds the beliefs described in Equations 1 through 6.

2. *Proof that the wary equilibrium exists.* Suppose the Receiver plays $a_R(2) = 2$ and $a_R(3) = 1$: they accept a recommendation of 2 and reject a recommendation of 3. The aligned-type Sender’s unique best response is $r(1) = 1$ and $r(2) = r(3) = 2$, since they want the Receiver’s action to be as close as possible to $\hat{a}(\theta)$. The uniqueness of the best response for $\theta = 1$ arises from the assumption that the aligned-type Sender recommends the action closest to $\hat{a}(\theta)$ when indifferent between recommendations. If the aligned-type Sender uses this strategy, the Receiver forms the following beliefs:

$$\mu(\theta, C|3) = q_\theta \text{ for all } \theta \quad (8)$$

$$\mu(\theta, A|3) = 0 \text{ for all } \theta \quad (9)$$

$$\mu(1, A|2) = 0 \quad (10)$$

$$\mu(2, A|2) = \frac{q_2}{q_2 + q_3} \quad (11)$$

$$\mu(3, A|2) = \frac{q_3}{q_2 + q_3} \quad (12)$$

$$\mu(1, A|1) = 1 \quad (13)$$

$$\mu(\theta, A|1) = 0 \text{ for all } \theta \neq 1 \quad (14)$$

According to these beliefs, accepting a recommendation of 2 will always bring the Receiver

closer to $\hat{a}(\theta)$ than will rejecting it (i.e., sticking with their status quo). Thus, it is always best for the Receiver to accept a recommendation of 2. Rejecting a recommendation of 3 is better than accepting it if

$$q_1 u(2\alpha) + q_2 u(\alpha) + q_3 u(0) \leq q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha)$$

which simplifies to

$$q_3 \leq q_1 \tag{15}$$

Thus, if Equation 15 holds, there exists a “wary” pure-strategy weak perfect Bayesian equilibrium where the aligned-type Sender plays $r(3) = 2$ and $r(\theta) = \hat{a}(\theta)$ for all $\theta \neq 3$, the Receiver plays $a_R(2) = 2$ and $a_R(3) = 1$, and the Receiver holds the beliefs described in Equations 8 through 14.

3. Proof that no other pure-strategy equilibria exist. The proofs of the existence of the trusting and wary equilibria show that the aligned-type Sender has a unique best response to each Receiver strategy that involves accepting a recommendation of 2.¹ To show that the trusting and wary equilibria are the unique equilibria of the recommendation game, I therefore prove that there can be no equilibrium where the Receiver rejects a recommendation of 2.

Suppose the Receiver plays a strategy that involves rejecting a recommendation of 2. The aligned-type Sender’s unique best response is $r(\theta) = \hat{a}(\theta)$ for all θ , given that they want the Receiver’s action to be as close as possible to $\hat{a}(\theta)$ and recommend the action closest to $\hat{a}(\theta)$ when indifferent² between recommendations. If the aligned-type Sender uses this strategy, the Receiver would believe $\mu(2, A|2) = 1$, $\mu(\theta, A|2) = 0$ for all $\theta \neq 2$. Stated otherwise, receiving a recommendation of 2 indicates that the state is 2. The Receiver could thus profitably deviate to accepting a recommendation of 2. Thus, there is no weak perfect Bayesian equilibrium where the Receiver rejects a recommendation of 2. As a result, the trusting and wary equilibria are the unique pure-strategy weak perfect Bayesian equilibria of the recommendation game. ■

A.1.2 Proof of Lemma 1

I will separately consider the Receiver’s best response to each aligned-type Sender strategy under the proposed parameter restriction.

¹Recall that the Receiver has two strategies that involve accepting a recommendation of 2: accept all (i.e., accept 2, accept 3) and accept-reject (i.e., accept 2, reject 3).

²When the Receiver rejects a recommendation of 2, in state 2, the aligned-type Sender is indifferent between all possible recommendations. When the Receiver rejects all recommendations, the aligned-type Sender is additionally indifferent between all possible recommendations in states 1 and 3.

1. *Best response to full truth-telling.*

- Best response to $r = 2$. Under full truth-telling, the aligned-type Sender only ever recommends 2 in state 2. Since the commitment-type Sender never recommends 2, upon observing $r = 2$, the Receiver knows with certainty that $\theta = 2$. The Receiver's expected utility of accepting a recommendation of 2 is therefore $u(0)$, while their expected utility of rejecting a recommendation of 2 is $u(\alpha)$. Since $u'(\cdot) < 0$, we have $u(0) > u(\alpha)$. Thus, 2 is the Receiver's unique best response to $r = 2$.
- Best response to $r = 3$. Under full truth-telling, the aligned-type Sender recommends 3 in state 3. Since the commitment-type Sender recommends 3 in all states, the Receiver strictly prefers rejecting a recommendation of 3 over accepting it if:

$$\begin{aligned} \Pr(\theta = 1|r = 3)u(0) + \Pr(\theta = 3|r = 3)u(2\alpha) &> \Pr(\theta = 1|r = 3)u(2\alpha) + \Pr(\theta = 3|r = 3)u(0) \\ [\Pr(\theta = 1|r = 3) - \Pr(\theta = 3|r = 3)][u(0) - u(2\alpha)] &> 0 \\ [q_1(1 - p) - q_3][u(0) - u(2\alpha)] &> 0 \end{aligned} \tag{16}$$

Since $u'(\cdot) < 0$, we have $u(0) > u(2\alpha)$. For Equation 16 to hold, we must then have $q_1(1 - p) > q_3$, which is the proposed parameter restriction. Thus, 1 is the Receiver's unique best response to $r = 3$.

Combining these results, if $q_1(1 - p) > q_3$ and the Receiver believes the Sender is playing full truth-telling, then accept-reject ($a_R(2) = 2, a_R(3) = 1$) is their unique optimal strategy.

2. *Best response to partial truth-telling.*

- Best response to $r = 2$. Under partial truth-telling, the aligned-type Sender recommends 2 in states 2 and 3. Since the commitment type never recommends 2, after receiving a recommendation of 2, the Receiver forms the following beliefs:

$$\begin{aligned} \Pr(\theta = 2|r = 2) &= \frac{q_2}{q_2 + q_3} \\ \Pr(\theta = 3|r = 2) &= \frac{q_3}{q_2 + q_3} \end{aligned}$$

According to these beliefs, accepting a recommendation of 2 will always bring the Receiver closer to $\hat{a}(\theta)$ than will rejecting it (i.e., sticking with their status quo). Thus, 2 is the Receiver's unique best response to $r = 2$.

- Best response to $r = 3$. Under partial truth-telling, the aligned-type Sender never recommends 3. Since the commitment-type Sender recommends 3 in all states, the Receiver strictly prefers rejecting a recommendation of 3 over accepting it if:

$$\begin{aligned} q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha) &> q_1 u(2\alpha) + q_2 u(\alpha) + q_3 u(0) \\ q_1 &> q_3 \end{aligned} \tag{17}$$

Equation 17 is directly implied by the proposed parameter restriction. Thus, 1 is the Receiver's unique best response to $r = 3$.

Combining these results, if $q_1(1-p) > q_3$ and the Receiver believes the Sender is playing partial truth-telling, then accept-reject ($a_R(2) = 2, a_R(3) = 1$) is their unique optimal strategy. ■

A.2 Existence of a Mixed-Strategy Equilibrium

PROPOSITION 2: *When $q_1 \geq q_3 \geq q_1(1-p)$, the recommendation game has a mixed-strategy weak perfect Bayesian equilibrium where...*

- *The aligned-type Sender plays $r(1) = 1, r(2) = 2$, and mixes in state 3: they play $r = 3$ with probability π^* and $r = 2$ with probability $1 - \pi^*$, where $\pi^* = \frac{(1-p)(q_1-q_3)}{pq_3}$.*
- *The Receiver plays $a_R(2) = 2$ and mixes when $r = 3$: they play $a_R(3) = 3$ with probability σ^* and $a_R(3) = 1$ with probability $1 - \sigma^*$, where $\sigma^* = \frac{u(\alpha)-u(2\alpha)}{u(0)-u(2\alpha)}$.*

Furthermore, the equilibrium described above is the unique mixed-strategy weak perfect Bayesian equilibrium of the recommendation game.

Proof of Proposition 2. I begin by verifying the indifference conditions for each player. Next, I ensure that neither has a profitable deviation from the proposed equilibrium strategy profile. Finally, I show that no other mixed-strategy weak perfect Bayesian equilibria exist.

1. *Verifying indifference conditions.* I focus first on the Receiver. Given the aligned-type Sender's proposed mixed strategy, the Receiver forms the following beliefs:

$$\mu(1, C|3) = \frac{q_1(1-p)}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} \tag{18}$$

$$\mu(2, C|3) = \frac{q_2(1-p)}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} \tag{19}$$

$$\mu(3, C|3) = \frac{q_3(1-p)}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} \tag{20}$$

$$\mu(3, A|3) = \frac{q_3 p \pi^*}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} \tag{21}$$

$$\mu(1, A|1) = 1 \quad (22)$$

$$\mu(2, A|2) = \frac{q_2}{q_2 + q_3(1 - \pi^*)} \quad (23)$$

$$\mu(3, A|2) = \frac{q_3(1 - \pi^*)}{q_2 + q_3(1 - \pi^*)} \quad (24)$$

$$\mu(\theta, A|r) = 0 \text{ for } (\theta, r) \in \{(2, 1), (3, 1), (1, 2), (1, 3)\} \quad (25)$$

Given these beliefs, for the Receiver to mix between $a_R(3) = 3$ and $a_R(3) = 1$, we must have

$$\frac{1}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} (q_1(1-p)u(2\alpha) + q_2(1-p)u(\alpha) + (1-p + p\pi^*)q_3 u(0)) =$$

$$\frac{1}{(1-p)(q_1 + q_2 + q_3) + q_3 p \pi^*} (q_1(1-p)u(0) + q_2(1-p)u(\alpha) + (1-p + p\pi^*)q_3 u(2\alpha))$$

which simplifies to

$$[q_3(1-p + p\pi^*) - q_1(1-p)][u(0) - u(2\alpha)] = 0 \quad (26)$$

Since $u(0) > u(2\alpha)$, for Equation 26 to hold, we must have

$$q_3(1-p + p\pi^*) = q_1(1-p) \quad (27)$$

Solving Equation 27 for π^* yields

$$\pi^* = \frac{(q_1 - q_3)(1-p)}{q_3 p} \quad (28)$$

To ensure $\pi^* \geq 0$, we must have $q_1 \geq q_3$. To ensure $\pi^* \leq 1$, we must have $q_1(1-p) \leq q_3$.

I turn now to the aligned-type Sender. Since the Receiver accepts a recommendation of 2 and mixes between accepting and rejecting a recommendation of 3, for the aligned-type Sender to mix between $r(3) = 3$ and $r(3) = 2$,³ we must have:

$$u(\alpha) = \sigma^* u(0) + (1 - \sigma^*) u(2\alpha)$$

which simplifies to

$$\sigma^* = \frac{u(\alpha) - u(2\alpha)}{u(0) - u(2\alpha)} \quad (29)$$

³Playing $r(3) = 1$ is strictly dominated: since the Receiver takes action 1 in that case, the aligned-type Sender's utility would be $u(2\alpha)$, which is strictly lower than their utility from recommending either 2 or 3 in state 3.

Since $u(0) > u(\alpha) > u(2\alpha)$ by the properties of the utility function, $\sigma^* \in (0, 1)$.

2. *Ensuring no profitable deviations.* As argued below, neither player has any profitable deviations from this mixed strategy profile.

- Aligned-type Sender, $\theta = 2$. Since the Receiver accepts a recommendation of 2, $r(2) = 2$ yields $u(0)$. Given the Receiver's mixed strategy, $r(2) = 3$ yields $\sigma^*u(\alpha) + (1 - \sigma^*)u(\alpha) = u(\alpha)$. Since the Receiver takes action 1 if it is recommended to them, $r(2) = 1$ also yields $u(\alpha)$. Since $u(0) > u(\alpha)$, there are no profitable deviations for the aligned-type Sender in state 2.
- Aligned-type Sender, $\theta = 1$. Since the Receiver takes action 1 if it is recommended to them, $r(1) = 1$ yields $u(0)$. Since the Receiver accepts a recommendation of 2, $r(2) = 2$ yields $u(\alpha)$. Given the Receiver's mixed strategy, $r(2) = 3$ yields $\sigma^*u(2\alpha) + (1 - \sigma^*)u(0)$. Since $\sigma^* \in (0, 1)$ and $u(0) > u(\alpha) > u(2\alpha)$, there are no profitable deviations for the aligned-type Sender in state 1.
- Receiver, $r = 2$. When the Receiver gets a recommendation of 2, they know that the state is either 2 or 3 (see Equations 23 through 25). Playing $a_R(2) = 2$ yields $q_2u(0) + q_3(1 - \pi^*)u(\alpha)$ while playing $a_R(2) = 1$ yields $q_2u(\alpha) + q_3(1 - \pi^*)u(2\alpha)$, which is strictly lower. Thus, the Receiver has no profitable deviation when they are recommended 2.

Therefore, when $q_1 \geq q_3 \geq q_1(1 - p)$, the recommendation game has a mixed-strategy weak perfect Bayesian equilibrium where the Receiver's beliefs are described by Equations 18 through 25, and:

- The aligned-type Sender plays $r(1) = 1, r(2) = 2$, and in state 3, plays $r = 3$ with probability π^* and $r = 2$ with probability $1 - \pi^*$, where $\pi^* = \frac{(1-p)(q_1-q_3)}{pq_3}$.
- The Receiver plays $a_R(2) = 2$, and when recommended c , plays $a_R(3) = 3$ with probability σ^* and $a_R(3) = 1$ with probability $1 - \sigma^*$, where $\sigma^* = \frac{u(\alpha)-u(2\alpha)}{u(0)-u(2\alpha)}$.

3. *Ensuring no other mixed-strategy equilibria exist.* As proven in Proposition 1, the Receiver always accepts $r = 2$ in equilibrium. Given this, the aligned-type Sender's unique best response in states 1 and 2 is to recommend the action corresponding to the state. Thus, mixing is only possible for the aligned-type Sender when $\theta = 3$, and for the Receiver when they receive $r = 3$. This proves that the mixed-strategy equilibrium identified above is the unique mixed-strategy weak perfect Bayesian equilibrium of the recommendation game. ■

A.3 Wary Equilibrium's Robustness to Alternative Assumptions

A.3.1 Receiver can take any action

Consider a variation of the model presented in Section 2 where the Receiver's action space is $a_R(r) \in \{1, 2, 3\}$ instead of $a_R(r) \in \{1, r\}$.

PROPOSITION A.3.1: Suppose $a_R(r) \in \{1, 2, 3\}$, $q_1 \geq q_3$, $q_2 \geq q_3$ and $q_1 \geq q_2 + q_3 \cdot \frac{u(\alpha) - u(2\alpha)}{u(0) - u(\alpha)}$. Then, the recommendation game has a wary equilibrium.

First, I define and prove a result that will help to interpret the proof of Proposition A.3.1.

LEMMA 2: When $q_1 \geq q_3$ and $q_1 \geq q_2 + q_3 \cdot \frac{u(\alpha) - u(2\alpha)}{u(0) - u(\alpha)}$, action 1 is the Receiver's ex-ante optimal action.

Proof of Lemma 2. For action 1 to be the Receiver's ex-ante optimal action, its ex-ante expected payoff must be weakly greater than that of actions 2 and 3. That is, the following conditions must hold:

$$q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha) \geq (q_1 + q_3) u(\alpha) + q_2 u(0) \quad (30)$$

$$q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha) \geq q_3 u(0) + q_2 u(\alpha) + q_1 u(2\alpha) \quad (31)$$

Equations 30 and 31 simplify to the following conditions, respectively:

$$q_1 \geq q_2 + q_3 \cdot \frac{u(\alpha) - u(2\alpha)}{u(0) - u(\alpha)} \quad (32)$$

$$q_1 \geq q_3 \quad (33)$$

Thus, when Equations 32 and 33 hold, 1 is the Receiver's ex-ante optimal action. ■

Proof of Proposition A.3.1. Suppose the Receiver plays $a_R(1) = 1$, $a_R(2) = 2$ and $a_R(3) = 1$. This strategy is a natural extension of the Receiver's accept-reject strategy in a setting where they can choose any action, and action 1 is optimal under their prior. The aligned-type Sender's unique⁴ best response is their partial truth-telling strategy; that is, $r(1) = 1$ and $r(2) = r(3) = 2$.

If the aligned-type Sender plays their partial truth-telling strategy, the Receiver forms the

⁴The uniqueness of the best response arises from the assumption that the aligned-type Sender recommends the action closest to $\hat{a}(\theta)$ when indifferent between recommendations.

beliefs outlined in Equations 8 through 14. To complete the proof, it must therefore be shown that it is optimal for the Receiver to play $a_R(1) = 1$, $a_R(2) = 2$ and $a_R(3) = 1$, given their beliefs. Since the Receiver knows the state is 1 if they are recommended 1, it is always optimal for them to play $a_R(1) = 1$. If the Receiver is recommended 2, they know the state is either 2 or 3. From the beliefs outlined in Equations 11 and 12, it is easy to see that $a_R(2) = 2$ is optimal if $q_2 \geq q_3$. Finally, a recommendation of 3 reveals no new information about the true state. In order for $a_R(3) = 1$ to be optimal, it must yield a greater expected payoff than $a_R(3) = 3$ and $a_R(3) = 2$, which is true under the following conditions:

$$(1 - p)[q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha)] \geq (1 - p)[q_1 u(2\alpha) + q_2 u(\alpha) + q_3 u(0)] \quad (34)$$

$$(1 - p)[q_1 u(0) + q_2 u(\alpha) + q_3 u(2\alpha)] \geq (1 - p)[q_1 u(\alpha) + q_2 u(0) + q_3 u(\alpha)] \quad (35)$$

Equation 34 simplifies to $q_1 \geq q_3$, and Equation 35 simplifies to $q_1 \geq q_2 + q_3 \cdot \frac{u(\alpha) - u(2\alpha)}{u(0) - u(\alpha)}$. Per Lemma 2, these conditions imply that action 1 is ex-ante optimal for the Receiver. Thus, when the Receiver can take any action, the aligned-type Sender can play a partial truth-telling strategy in equilibrium if $q_1 \geq q_3$, $q_2 \geq q_3$ and $q_1 \geq q_2 + q_3 \cdot \frac{u(\alpha) - u(2\alpha)}{u(0) - u(\alpha)}$. ■

A.3.2 Non-equidistant states

Consider a variation of the model presented in Section 2 where, instead of there being distance 1 between adjacent states, there is distance d_{12} between states 1 and 2 and d_{23} between states 2 and 3, where $d_{12}, d_{23} \geq 0$.

PROPOSITION A.3.2: Suppose there is distance d_{12} between states 1 and 2 and d_{23} between states 2 and 3, where $d_{12}, d_{23} \geq 0$. Suppose further that $q_1 \geq q_2 \frac{u(\alpha d_{23}) - u(\alpha d_{12})}{u(0) - u(\alpha(d_{12} + d_{23}))} + q_3$. Then, the recommendation game has a wary equilibrium.

Proof of Proposition A.3.2. Suppose the Receiver plays their accept-reject strategy; that is, $a_R(2) = 2$ and $a_R(3) = 1$. The aligned-type Sender's unique best response is their partial truth-telling strategy; that is, $r(1) = 1$ and $r(2) = r(3) = 2$.

Now, suppose the aligned-type Sender plays their partial truth-telling strategy. The Receiver then forms the beliefs outlined in Equations 8 through 14. I now show that, under the conditions of the Proposition, it is optimal for the Receiver to play accept-reject, given their beliefs. According to these beliefs, accepting a recommendation of 2 will bring the Receiver closer to the action corresponding to the true state than will rejecting it. Thus, it is always optimal for the Receiver to accept a recommendation of 2. It is optimal for the Receiver to

reject a recommendation of 3 if

$$(1-p)[q_1u(0) + q_2u(\alpha d_{12}) + q_3u(\alpha(d_{12} + d_{23}))] \geq (1-p)[q_1u(\alpha(d_{12} + d_{23})) + q_2u(\alpha d_{23}) + q_3u(0)]$$

which simplifies to

$$q_1 \geq q_2 \frac{u(\alpha d_{23}) - u(\alpha d_{12})}{u(0) - u(\alpha(d_{12} + d_{23}))} + q_3 \quad (36)$$

Thus, the recommendation game has a wary equilibrium if Equation 36 is satisfied. ■

A.3.3 Strategic self-interested Sender

Consider a variation of the model presented in Section 2 where the non-strategic commitment-type Sender is instead a strategic, self-interested player. This player, who I will call the selfish-type Sender, either (1) receives a bonus if the Receiver takes action 3, or (2) places some weight on each of two outcomes: maximizing the Receiver's wellbeing and recommending action 3. The selfish-type Sender's preferences are described by one of the following payoff functions:

$$U_1(a_R(r)) = \begin{cases} \pi & \text{if } a_R(r) \neq 3 \\ \pi + B & \text{if } a_R(r) = 3, \text{ where } B > 0 \end{cases}$$

$$U_2(a_R(r)) = \gamma \cdot u(\alpha|a_R(r) - \hat{a}(\theta)|) + (1 - \gamma) \cdot B \cdot \mathbb{1}\{r = 3\}, \text{ where } B > 0 \text{ and } \gamma \in [0, 1]$$

$U_1(\cdot)$ captures the preferences of a Sender who stands to benefit significantly more when the Receiver takes one particular action, while $U_2(\cdot)$ captures the preferences of a Sender who is pressured to recommend certain actions.

PROPOSITION A.3.3:

- (i) Suppose the selfish-type Sender's preferences are described by $U_1(\cdot)$. Then, the recommendation game has a wary equilibrium.
- (ii) Suppose the selfish-type Sender's preferences are described by $U_2(\cdot)$, and $\frac{B}{u(0) - u(\alpha) + B} \geq \gamma$. Then, the recommendation game has a wary equilibrium.

Proof of Proposition A.3.3. I wish to show that, for each of the selfish-type Sender's possible payoff functions, the recommendation game has an equilibrium where the Receiver plays accept-reject, the aligned-type Sender plays their partial truth-telling strategy, and the selfish-

type Sender always recommends action 3. If each Sender type uses these strategies, the Receiver forms the beliefs outlined in Equations 8 through 14. As shown in the proof of Proposition 1, given these beliefs, it is optimal for the Receiver to accept a recommendation of 2 and reject a recommendation of 3.

I next check that the strategy of each of the Sender's types is a best response to the Receiver's strategy. As shown in the proof of Proposition 1, the aligned-type Sender's unique best response to the Receiver's accept-reject strategy is their partial truth-telling strategy. When the selfish-type Sender has the preferences described by $U_1(\cdot)$, they are indifferent between the Receiver accepting a recommendation of 2 and rejecting any given recommendation. Thus, given the Receiver's strategy, a selfish-type Sender with these preferences cannot profitably deviate from recommending 3 in all states. As a result, when the selfish-type Sender has the preferences described by $U_1(\cdot)$, the recommendation game always has a wary equilibrium.

When the selfish-type Sender has the preferences described by $U_2(\cdot)$, they may have a profitable deviation if recommending an action other than 3 leads the Receiver to take an action that is closer to the state. This can only be the case in state 2, where recommending 2 leads the Receiver to take the action corresponding to the state. The selfish-type Sender prefers recommending 3 in state 2 if

$$\gamma \cdot u(\alpha) + (1 - \gamma)B \geq \gamma \cdot u(0)$$

which simplifies to

$$\frac{B}{u(0) - u(\alpha) + B} \geq \gamma \quad (37)$$

Intuitively, the selfish-type Sender must place a sufficiently high weight on their bonus, relative to the weight they place on the Receiver's wellbeing. Thus, when the selfish-type Sender has the preferences described by $U_2(\cdot)$, the recommendation game has a wary equilibrium if Equation 37 is satisfied. ■

A.3.4 Alternative status quo and commitment actions

In the model presented in Section 2, it was assumed that the Receiver's status quo and the action the commitment type recommends are located at opposite boundaries of the action space. Proposition A.3.4 shows that this assumption is necessary for the existence of a wary equilibrium.

PROPOSITION A.3.4: The existence of a wary equilibrium requires $|s - c| = 2$.

Proof of Proposition A.3.4. In a wary equilibrium, the Receiver rejects the commitment recommendation, and, in the state corresponding to the commitment recommendation, the aligned-type Sender responds by recommending a non-commitment action that the Receiver accepts. I proceed by showing that, if $|s - c| \neq 2$, these strategies cannot be part of a pure-strategy weak perfect Bayesian equilibrium.

Suppose that, instead of the status quo and commitment actions being distance 2 apart, they were distance 1 apart. Specifically, without loss of generality, assume the commitment type recommends 2 while the status quo remains 1. The Receiver's wary equilibrium strategy would then be to reject a recommendation of 2 and accept a recommendation of 3. For a wary equilibrium to exist, the aligned-type Sender's strategy must then involve recommending an action other than 2 in state 2. However, regardless of the action they recommend in state 2, the aligned-type Sender receives utility $u(\alpha)$.⁵ Per the assumption that the aligned-type Sender recommends the action closest to $\hat{a}(\theta)$ when indifferent between recommendations, they then play $r(2) = 2$; that is, they truthfully reveal the state when it corresponds to the commitment recommendation. This contradicts the existence of a wary equilibrium. ■

⁵If they recommend 1 (the status quo), the Receiver takes action 1. If they recommend 2, the Receiver rejects and takes action 1. If they recommend 3, the Receiver accepts and takes action 3. In all cases, the action the Receiver takes is distance 1 from the action corresponding to the state.

B Experiment Instructions

In the experiment, Receivers were called “Choosers” and aligned-type Senders were called “Advisors”. Any instructions or tasks that are specific to a particular role (Chooser or Advisor) or a particular treatment (Main or Beliefs) are indicated by text boxes.

Welcome to the experiment!

General instructions. This is an experiment designed to study decision-making. If you pay close attention to the instructions, you can earn a significant amount of money. Please ensure all of your electronics (cell phones, smart watches, etc.) are put away, and do not talk with others during the experiment. If you have a question, please raise your hand and an experimenter will come answer it in private.

MAIN TREATMENT ONLY

Structure of the experiment. The experiment consists of multiple rounds of a game, followed by a short survey. You will receive detailed instructions prior to each part of the experiment.

BELIEFS TREATMENT ONLY

Structure of the experiment. The experiment consists of multiple rounds of a game. Before several of those rounds, you will complete a guessing task. Finally, you will complete a short survey. You will receive detailed instructions prior to each part of the experiment.

Payment. Your total earnings from the experiment will consist of several components.

- *Show-up fee.* You will receive a \$5.00 payment for completing the experiment.
- *Quiz.* After reading the experiment’s instructions but before proceeding to the main task, you will complete a quiz that will test your understanding of the instructions. You will receive \$1.00 if you answer all quiz questions correctly on your first try.

MAIN TREATMENT ONLY

- *Randomly-selected round of the game.* One round of the game has been randomly selected to count for your payment.

BELIEFS TREATMENT ONLY

- *Randomly-selected round and guessing task.* One round of the game, as well as one guessing task, have each been randomly selected to count for your payment.

The Plant Pot Game - Instructions

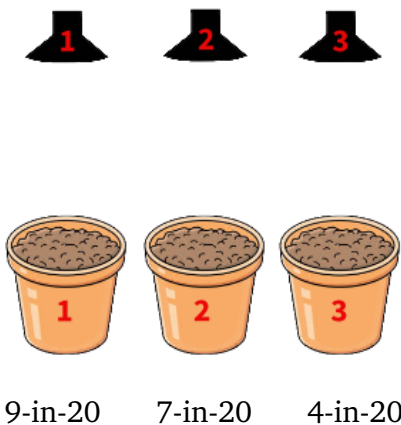
There are three pots, each with a sprinkler above it. The “plant pot” has a plant seed in it; the other two do not.

CHOOSERS ONLY

You are playing the role of **Chooser**. You will not know for sure which pot is the plant pot, but you will know each pot’s chance of being the plant pot. Specifically, pot 1 has a 9-in-20 chance, pot 2 has a 7-in-20 chance and pot 3 has a 4-in-20 chance.

ADVISORS ONLY

Participants playing the role of Chooser know that pot 1 has a 9-in-20 chance of being the plant pot, pot 2 has a 7-in-20 chance and pot 3 has a 4-in-20 chance. You, however, are playing the role of **Advisor**. As we’ll explain later, Advisors have more information about the plant pot than Choosers do.



CHOOSERS ONLY

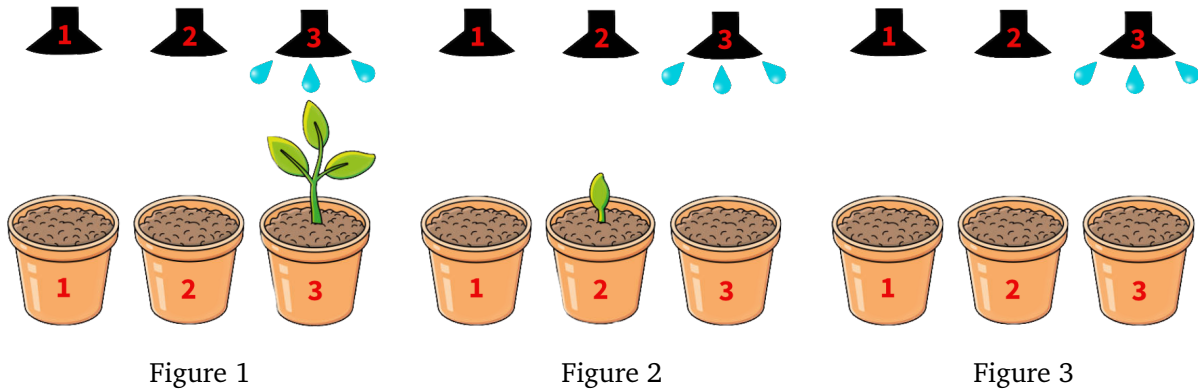
The pots, the sprinklers and each pot’s chance of being the plant pot.

ADVISORS ONLY

The pots, the sprinklers and the information Choosers have (each pot’s chance of being the plant pot).

When a sprinkler is turned on, it sprays water directly below it, as well as below to the right and left of it. The pot closest to it receives the most water; the pot farthest from it receives the

least. A seed will grow into a larger plant if it receives more water.



Example: Turning on sprinkler 3 will produce a large plant if pot 3 is the plant pot (as in Figure 1), a small plant if pot 2 is the plant pot (as in Figure 2), and no plant if pot 1 is the plant pot (as in Figure 3).

Your goal

CHOOSERS ONLY

As a Chooser, your goal is to turn on the sprinkler that's as close as possible to the plant pot, and thus to produce the largest possible plant.

ADVISORS ONLY

As an Advisor, you will be paired with a participant playing the role of Chooser. Choosers and Advisors have the same goal: for the Chooser to turn on the sprinkler that's as close as possible to the plant pot, and thus to produce the largest possible plant.

Your task

CHOOSERS ONLY

While sprinkler 1 will be turned on by default, you may receive a recommendation to switch to one of the other sprinklers. Your task requires you to consider two scenarios: the scenario where you receive a recommendation to switch to sprinkler 2, and the one where you receive a recommendation to switch to sprinkler 3. You must decide whether you will accept the proposed switch or stick with the default in each of these scenarios. If you are recommended to stick with the default, you will automatically do so.

ADVISORS ONLY

Your task requires you to consider three scenarios: the scenario where pot 1 is the plant pot, the one where pot 2 is the plant pot, and the one where pot 3 is the plant pot. You must choose a sprinkler to recommend to the Chooser in each scenario. Think of this task as if you know a specific pot (pot 1, 2 or 3) is the plant pot, and with this knowledge, must recommend a sprinkler to the Chooser.

Sprinkler 1 will be turned on by default; you may recommend that the Chooser either stick with this default or switch to one of the other sprinklers. While you are completing this task, the Chooser will decide whether, if recommended to switch to a given sprinkler, they will accept the proposed switch or stick with the default. If they are recommended to stick with the default, they will automatically do so.

Sources of recommendations

CHOOSERS ONLY

The recommendation you receive will be from one of two sources: an Advisor or the computer. An Advisor is another participant who has the same goal as you, and can recommend sticking with the default or switching to one of the other sprinklers. While you are completing your task, they will choose a sprinkler to recommend for each possible plant pot. You can think of the Advisor's task as if they know a specific pot (pot 1, 2 or 3) is the plant pot, and with this knowledge, must recommend you a sprinkler. The computer recommends to switch to sprinkler 3 regardless of which pot is the plant pot. There is a 1-in-3 chance that you will receive an Advisor's recommendation, and a 2-in-3 chance that you will receive the computer's recommendation.

ADVISORS ONLY

The recommendation the Chooser receives will be from one of two sources: you or the computer. The computer recommends to switch to sprinkler 3 regardless of which pot is the plant pot. There is a 1-in-3 chance that the Chooser will receive your recommendation, and a 2-in-3 chance that they will receive the computer's recommendation. When making their decisions, Choosers know their chance of receiving a recommendation from each potential source. They also know that Advisors have the same goal as them, and must choose a sprinkler to recommend for each possible plant pot.

Rounds

CHOOSERS ONLY

You will play 22 rounds of this game. At the beginning of each round, we will randomly re-match each Chooser with an Advisor. We will also determine the plant pot and source of the recommendation you will receive according to the chances mentioned earlier. After you and the Advisor you are paired with have made your decisions, the outcome of the game will be determined. At the end of each round, you will receive feedback about that outcome, as well as the decisions of the Advisor you were paired with.

ADVISORS ONLY

You will play 22 rounds of this game. At the beginning of each round, we will randomly re-match each Chooser with an Advisor. We will also determine the plant pot and source of the recommendation the Chooser will receive according to the chances mentioned earlier. After you and the Chooser you are paired with have made your decisions, the outcome of the game will be determined. At the end of each round, you will receive feedback about that outcome, as well as the decisions of the Chooser you were paired with.

Round 1 is a “practice round”. It is intended to familiarize you with your task, and is not eligible to determine your payment from the experiment. At the end of round 1, you will see an outline of the feedback you will receive at the end of the rest of the rounds.

BELIEFS TREATMENT ONLY

Guessing tasks

CHOOSERS ONLY

Before rounds 2, 8, 14 and 20, you will (1) make guesses about the most-common decisions Advisors have made in that round in previous sessions of the experiment, and (2) state which recommendation the computer sends when any given pot is the plant pot. Part (2) does not involve guessing; its purpose is to ensure you remember which recommendation the computer always sends.

ADVISORS ONLY

Before rounds 2, 8, 14 and 20, you will make guesses about the most-common decisions Choosers have made in that round in previous sessions of the experiment.

Your payment

CHOOSERS ONLY

One round of the game has been randomly selected to count for your payment. You will receive \$16.00 if you produce a large plant in that round, \$13.00 if you produce a small plant, and \$10.00 if you produce no plant.

BELIEFS TREATMENT ONLY

One of the guessing tasks has also been randomly selected to count for your payment. You will receive \$5.00 if your guesses about Advisors' recommendations, as well as your statements about the recommendation the computer sends, are correct.

ADVISORS ONLY

One round of the game has been randomly selected to count for your payment. If the Chooser you are paired with receives your recommendation in that round, you will receive \$16.00 if they produce a large plant, \$13.00 if they produce a small plant, and \$10.00 if they produce no plant. If the Chooser you are paired with receives the computer's recommendation in that round, you will receive \$10.00 regardless of the size of the plant they produce.

BELIEFS TREATMENT ONLY

One of the guessing tasks has also been randomly selected to count for your payment. You will receive \$5.00 if your guesses are correct.

Quiz

On the next page, you will complete a quiz that tests your understanding of these instructions. You must answer all quiz questions correctly before proceeding to the first round. You will receive \$1.00 if you answer all quiz questions correctly on your first try. You will not be able to return to the instructions while completing the quiz, so please ensure you have read them carefully.

Quiz

Question 1. When a sprinkler is turned on...

- ☐ only the pot closest to it receives any water.
- ☐ the pot closest to it receives the most water; the pot farthest from it receives the least.

Question 2. The computer always recommends to...

- ☐ stick with sprinkler 1.
- ☐ switch to sprinkler 2.
- ☐ switch to sprinkler 3.

CHOOSERS ONLY

Question 3. Which of these statements is true?

- ☐ Advisors have the same goal as you.
- ☐ Advisors choose a sprinkler to recommend for each possible plant pot.
- ☐ Advisors can recommend sticking with the default or switching to one of the other sprinklers.

Question 4. In each round, you will be...

- ☐ randomly matched to an Advisor.
- ☐ matched with the same Advisor.

ADVISORS ONLY

Question 3. After receiving a recommendation to switch sprinklers, a Chooser can turn on...

- ☐ the default sprinkler or the recommended sprinkler.
- ☐ any sprinkler.

Question 4. Which of these statements is true?

- ☐ When making their decisions, Choosers know their chance of receiving a recommendation from each potential source.
- ☐ Choosers know that Advisors have the same goal as them.
- ☐ Choosers know that Advisors choose a sprinkler to recommend for each possible plant pot.
- ☐ All of the above statements are true.

Question 5. In each round, you will be...

- ☐ randomly matched to a Chooser.
- ☐ matched with the same Chooser.

Question 6. Your task consists of three choices: you must choose a sprinkler to recommend to the Chooser for each possible plant pot (pot 1, 2 or 3). You should think of this task as if...

- ☐ you know a specific pot (pot 1, 2 or 3) is the plant pot, and with this knowledge, must recommend a sprinkler.
- ☐ you are sending recommendations without knowing which pot is the plant pot.

Quiz complete

Congratulations! You have answered all of the quiz questions correctly. Once all participants have done so, we will proceed to the practice round.

Round 1 (practice round)

CHOOSERS ONLY

A new round has begun. You have been randomly matched to an Advisor, and the plant pot and source of the recommendation shown to you have been determined.

ADVISORS ONLY

A new round has begun. You have been randomly matched to a Chooser, and the plant pot and source of the recommendation shown to that Chooser have been determined.

Round 1 (practice round)

CHOOSERS ONLY

Summary of instructions

- A seed will grow into a larger plant if it receives more water.
- Your goal (same as the Advisor's): Produce the largest possible plant.
- Your task: Decide whether, if recommended to switch to a given sprinkler, you will accept the proposed switch or stick with the default (sprinkler 1).
- There is a 1-in-3 chance you will receive the Advisor's recommendation, and a 2-in-3 chance you will receive the computer's recommendation. The Advisor chooses a sprinkler to recommend for each possible plant pot. The computer always recommends to switch to sprinkler 3.



9-in-20 7-in-20 4-in-20

*The pots, the sprinklers and each pot's
chance of being the plant pot.*

Your task

Please decide whether you will accept or reject the recommendation in each of the scenarios below.

- You receive a recommendation to switch to SPRINKLER 2. Will you accept or reject this recommendation?
 - ☐ Accept (switch to sprinkler 2)
 - ☐ Reject (stick with sprinkler 1)
- You receive a recommendation to switch to SPRINKLER 3. Will you accept or reject

this recommendation?

- ☐ Accept (switch to sprinkler 3)
- ☐ Reject (stick with sprinkler 1)

ADVISORS ONLY

Summary of instructions

- A seed will grow into a larger plant if it receives more water.
- Your goal (same as the Chooser's): Produce the largest possible plant.
- Your task: Choose a sprinkler to recommend to the Chooser for each possible plant pot. The Chooser decides whether, if recommended to switch to a given sprinkler, they will accept the proposed switch or stick with the default (sprinkler 1).
- There is a 1-in-3 chance the Chooser will receive your recommendation, and a 2-in-3 chance they will receive the computer's recommendation. The computer always recommends to switch to sprinkler 3.



9-in-20 7-in-20 4-in-20

*The pots, the sprinklers and the information Choosers
have (each pot's chance of being the plant pot)*

Your task

Please choose a recommendation to send the Chooser in each of the scenarios below.

- You know that POT 1 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- You know that POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- You know that POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

Round 1 (practice round)

Feedback

Here is an outline of the feedback you will receive at the end of a round.

CHOOSERS ONLY

Advisor's decisions: The Advisor you were paired with made the following decisions.

- If Pot 1 is the plant pot, recommend to *[stick with sprinkler 1/switch to sprinkler 2/switch to sprinkler 3]*
- If Pot 2 is the plant pot, recommend to *[stick with sprinkler 1/switch to sprinkler 2/switch to sprinkler 3]*
- If Pot 3 is the plant pot, recommend to *[stick with sprinkler 1/switch to sprinkler 2/switch to sprinkler 3]*



This image will additionally show the sprinkler that was turned on and the plant that was produced.

Outcome of this round: *Here, you will receive feedback on...*

- *which pot was the plant pot*
- *the source of the recommendation you received*
- *the sprinkler that was turned on*
- *the size of the plant produced*

ADVISORS ONLY

Chooser's decisions: The Chooser you were paired with made the following decisions.

- If recommended to switch to sprinkler 2, *[accept/reject]* this recommendation
- If recommended to switch to sprinkler 3, *[accept/reject]* this recommendation

Outcome of this round: *Here, you will receive feedback on...*

- *which pot was the plant pot*
- *the source of the recommendation the Chooser received*
- *the sprinkler that was turned on*
- *the size of the plant produced*



This image will additionally show the sprinkler that was turned on and the plant that was produced.

Practice round complete

Congratulations! You have completed the practice round. Once all participants have done so, we will proceed to the rest of the experiment.

MAIN TREATMENT ONLY

Recall that one of the rounds that follow has been randomly selected to count for your payment. You should therefore treat each round as if it is the one that counts.

BELIEFS TREATMENT ONLY

Recall that one of the rounds that follow, as well as one of the guessing tasks, have been randomly selected to count for your payment. You should therefore treat each round and guessing task as if it is the one that counts.

AUTHOR'S NOTE: Participants then played 21 more rounds of the Plant Pot Game. In rounds 2-21, the introduction and decision pages were the same as in the practice round. The structure of the feedback page was the same in rounds 2-22, but its content depended on the outcome of the round. Below are examples of the feedback that players in each role may receive.

Round n

Feedback

CHOOSERS ONLY

Advisor's decisions: The Advisor you were paired with made the following decisions.

- If Pot 1 is the plant pot, recommend to stick with sprinkler 1
- If Pot 2 is the plant pot, recommend to switch to sprinkler 2
- If Pot 3 is the plant pot, recommend to switch to sprinkler 3



Outcome of this round:

- Pot 2 was the plant pot.
- You received the Advisor's recommendation, which, as described above, was to switch to sprinkler 2.

- You accepted this recommendation, thereby switching to sprinkler 2.
- As a result, you produced a large plant.

ADVISORS ONLY

Chooser's decisions: The Chooser you were paired with made the following decisions.

- If recommended to switch to sprinkler 2, accept this recommendation
- If recommended to switch to sprinkler 3, reject this recommendation



Outcome of this round:

- Pot 2 was the plant pot.
- The Chooser received your recommendation, which was to switch to sprinkler 2.
- As described above, the Chooser accepted this recommendation, thereby switching to sprinkler 2.
- As a result, the Chooser produced a large plant.

AUTHOR'S NOTE: In the beliefs treatment, participants viewed and completed the page below before the introduction page in rounds 2, 8, 14 and 20.

Guessing task for round m

CHOOSERS ONLY

Instructions

- **Your task:** We want you to think about the recommendations that tend to be sent in the round you are about to play, depending on which pot is the plant pot. You must (1) guess which recommendation has most-frequently been sent by Advisors when they knew a given pot was the plant pot in round m of previous sessions of this experiment, and (2) state which recommendation the computer sends when any given pot is the plant pot. Part (2) does not involve guessing; its purpose is to ensure you remember which recommendation the computer always sends.
- **Payment:** If this guessing task counts for your payment, you will be paid \$5.00 if all of your guesses about Advisors' recommendations, as well as all of your statements about the recommendation the computer sends, are correct.

Your task

(1) In ROUND m , which recommendation is most-frequently sent by **Advisors** when they know...

- ... POT 1 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... POT 2 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... POT 3 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

(2) In ALL ROUNDS, which recommendation does **the computer** send when...

- ... POT 1 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... POT 2 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... POT 3 is the plant pot?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

ADVISORS ONLY

Instructions

- **Your task:** We want you to think about the decisions Choosers tend to make in the round you are about to play. You must guess Choosers' most-common response to a recommendation to switch to a given sprinkler in round m of previous sessions of this experiment.
- **Payment:** If this guessing task counts for your payment, you will be paid \$5.00 if both of your guesses are correct.

Your task

In ROUND m , what is Choosers' most-common response to a recommendation to...

- ... switch to SPRINKLER 2?
 - ☐ Accept (switch to sprinkler 2)
 - ☐ Reject (stick with sprinkler 1)
- ... switch to SPRINKLER 3?
 - ☐ Accept (switch to sprinkler 3)
 - ☐ Reject (stick with sprinkler 1)

AUTHOR'S NOTE: In round 22, Choosers' introduction and decision pages were the same as in all previous rounds. These pages were different for Advisors (see below).

Round 22

AUTHOR'S NOTE: The two introduction sentences were the same as in all previous rounds.

*In this final round, you will be able to make different recommendations depending on (1) which pot is the plant pot **AND** (2) the Chooser's decision to accept or reject recommendations to switch sprinklers.*

Round 22

AUTHOR'S NOTE: The summary of the instructions was the same as in all previous rounds.

*In this final round, you will be able to make different recommendations depending on (1) which pot is the plant pot **AND** (2) the Chooser's decision to accept or reject recommendations to switch sprinklers.*

Your task

Please choose a recommendation to send the Chooser in each of the scenarios below.

You know the Chooser would ACCEPT to switch to SPRINKLER 2 and ACCEPT to switch to SPRINKLER 3, and...

- ... you know POT 1 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 3 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

You know the Chooser would ACCEPT to switch to SPRINKLER 2 and REJECT to switch to SPRINKLER 3, and...

- ... you know POT 1 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 3 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

You know the Chooser would REJECT to switch to SPRINKLER 2 and ACCEPT to switch to SPRINKLER 3, and...

- ... you know POT 1 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 3 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

You know the Chooser would REJECT to switch to SPRINKLER 2 and REJECT to switch to SPRINKLER 3, and...

- ... you know POT 1 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3
- ... you know POT 2 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

- ... you know POT 3 is the plant pot. What do you recommend to the Chooser?
 - ☐ Stick with sprinkler 1
 - ☐ Switch to sprinkler 2
 - ☐ Switch to sprinkler 3

MAIN TREATMENT ONLY

Rounds complete

Congratulations! You have now completed all rounds of the game. Click the “Next” button to proceed to the survey.

BELIEFS TREATMENT ONLY

Rounds and guessing tasks complete

Congratulations! You have now completed all rounds of the game and all of the guessing tasks. Click the "Next" button to proceed to the survey.

Survey (Page 1 of 2)

CHOOSERS ONLY

We’re interested in knowing more about how you decided whether to accept a recommendation.

ADVISORS ONLY

We’re interested in knowing more about how you decided which recommendations to send.

1. Did your approach to this decision remain constant, or did it change as the rounds progressed?
 - ☐ My approach remained constant.
 - ☐ My approach changed as the rounds progressed.
2. Please explain why your approach did or did not change as the rounds progressed. [Text field]
3. Do you have any other thoughts or comments on the experiment? If not, leave this field blank. [Text field]

Survey (Page 2 of 2)

We want to know more about you! Please take a moment to answer these demographic questions.

1. What degree are you currently pursuing?

- ☐ Bachelor's degree
- ☐ Master's degree
- ☐ Doctoral degree
- ☐ Other degree (not listed)

2. In what year did you begin the degree you are currently pursuing?

- ☐ 2024
- ☐ 2023
- ☐ 2022
- ☐ 2021
- ☐ 2020
- ☐ 2019
- ☐ 2018 or earlier

3. What is your primary field of study (e.g., Anthropology, Chemistry, Mechanical Engineering, etc.)? [Text field]

4. How old are you? [Text field]

5. What is your gender?

- ☐ Female
- ☐ Male
- ☐ Non-binary
- ☐ Other (not listed)
- ☐ Prefer not to answer

Your earnings (*example*)

MAIN TREATMENT ONLY

Round 10 was randomly selected to count for your payment. As summarized in the table below, your total earnings are **\$22.00**.

Earnings breakdown	
Show-up fee	\$5.00
Earnings from quiz	\$1.00
Earnings from round 10	\$16.00
<i>Total earnings</i>	<i>\$22.00</i>

BELIEFS TREATMENT ONLY

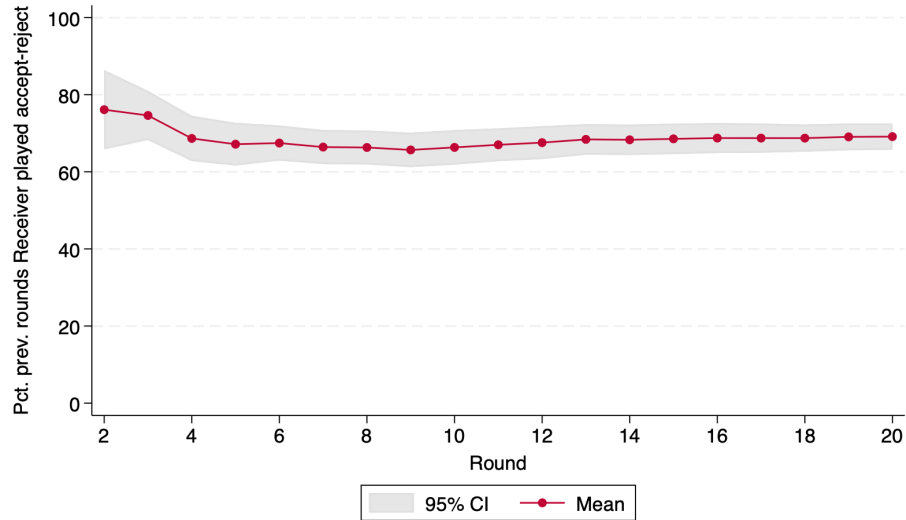
Round 10 and the guessing task for round 2 were randomly selected to count for your payment. As summarized in the table below, your total earnings are **\$27.00**.

Earnings breakdown	
Show-up fee	\$5.00
Earnings from quiz	\$1.00
Earnings from round 10	\$16.00
Earnings from guessing task for round 2	\$5.00
<i>Total earnings</i>	<i>\$27.00</i>

An experimenter will come by your desk shortly and pay you this amount in cash. Thanks for participating!

C Supplementary Analyses

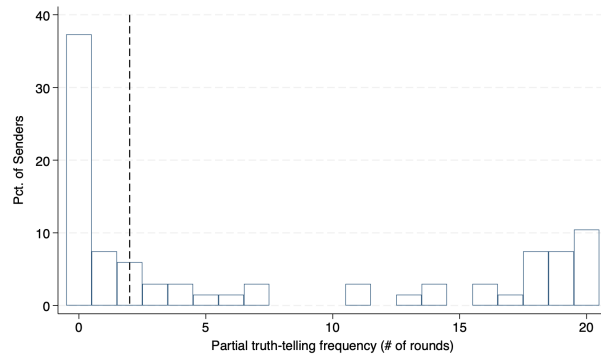
Figure C.1
Feedback Received by Senders



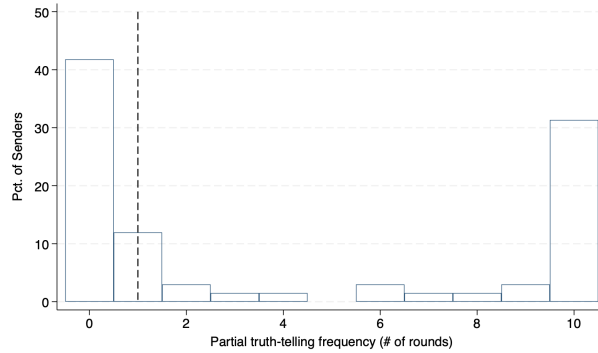
Notes. This figure summarizes the feedback Senders in the main treatment have received (on the y-axis) when choosing their strategy in round x (on the x-axis). The figure displays the percent of previous rounds in which Senders received the feedback that the Receiver they were paired with played accept-reject. The pink line indicates the mean (across Senders), and the grey area indicates the 95% confidence interval of that mean.

Figure C.2

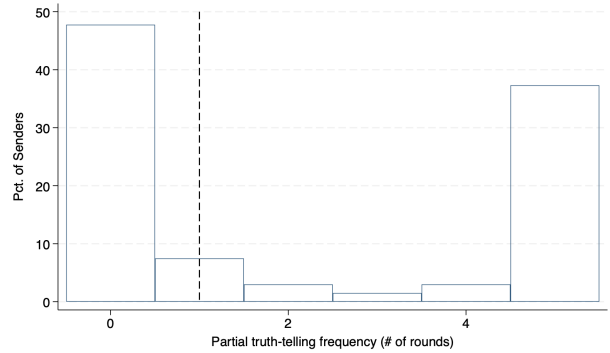
Senders' Optimal Strategy Play Frequency: Main Treatment, Player-Level



(a) All rounds



(b) Rounds 11 through 20



(c) Rounds 15 through 20

Notes. This figure displays the percent of Senders in the main treatment playing partial truth-telling in a given number of main rounds. Figure C.2a includes data from all 20 main rounds, Figure C.2b restricts the data to the final 10 main rounds, and Figure C.2c restricts the data to the final 5 main rounds. The dashed line indicates the median frequency.

Table C.1
Sender Strategy Type Classification (Main Treatment)

<i>Classification scheme</i>	Base	Mixed: ≥ 4 rds.	Mostly: ≥ 15 rds.	Both stricter criteria
<u><i>Strategy type</i></u>				
Mostly PTT	32.8%	32.8%	29.9%	29.9%
Mostly FTT	40.3%	40.3%	37.3%	37.3%
Mixed: PTT & FTT	4.5%	3.0%	9.0%	7.5%
Mostly other strategy	7.5%	7.5%	7.5%	7.5%
Mixed: PTT, FTT & other	14.9%	16.4%	16.4%	17.9%

Notes. Cells report the percentage of Senders of each type (“PTT” = partial truth-telling, “FTT” = full truth-telling). In the base specification, Senders are classified as (i) “mostly” using a strategy if they use it in at least 14 of 20 rounds, and (ii) mixed partial/full truth-tellers if they use partial and full truth-telling in at least 14 rounds combined, and use each of the two strategies in at least 3 rounds. Robustness specifications increase the threshold in (i) to 15 rounds, the threshold in (ii) to 4 rounds, or both.

Table C.2
Participant Characteristics

	Main	Beliefs	<i>p</i> -value
Age	21.1	21.3	0.95
% women	60.0	67.4	0.21
% bachelor's degree program	85.9	80.6	0.25
Comprehension quiz mistakes	0.99	1.15	0.21

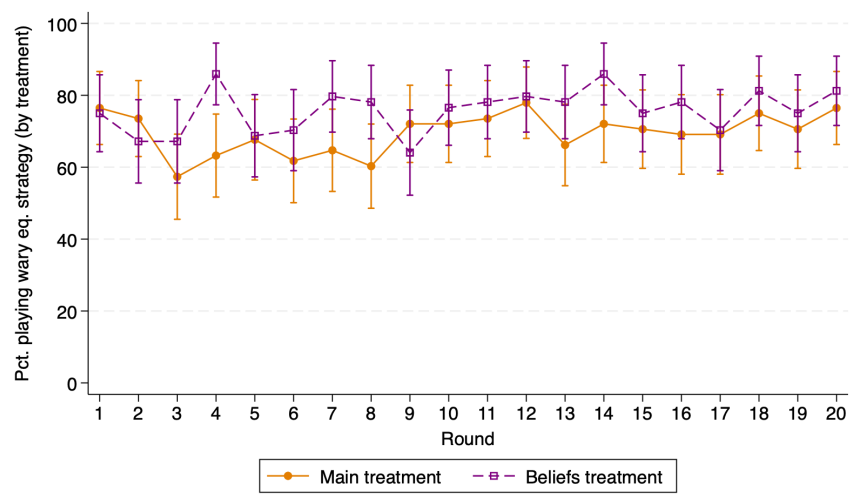
Notes. Mean participant characteristics by treatment. Excludes data of one participant from each treatment who withdrew from their session. The final column reports the *p*-value from a Wilcoxon-rank-sum test.

Table C.3
 Receivers' Beliefs about Senders' Strategies (Beliefs Treatment)

	Round 1	Round 7	Round 13	Round 19
Partial truth-telling	14.0	26.6	45.3	50.0
Full truth-telling	75.0	65.6	48.4	45.3
Other strategy	10.9	7.8	6.3	4.7

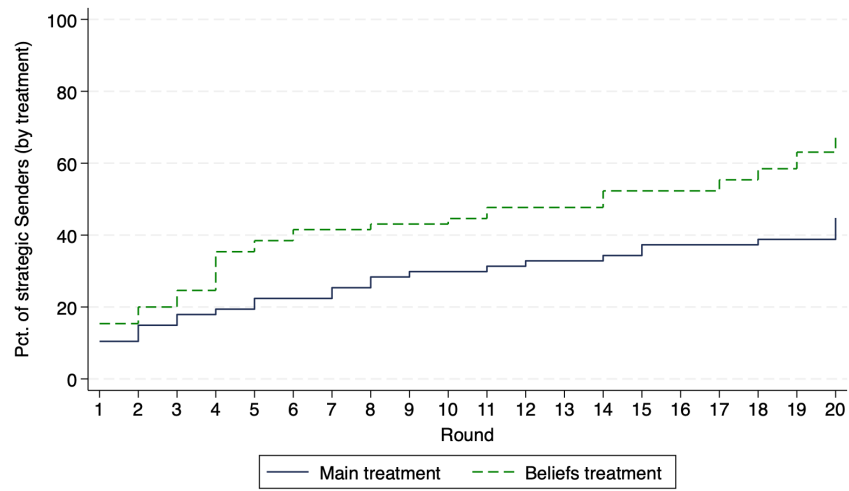
Notes. Percent of Receivers in the beliefs treatment who believe that a given strategy is the modal Sender strategy in a given round.

Figure C.3
Receivers' Frequency of Optimal Strategy Play (Round-Level), by Treatment



Notes. This figure displays the percent of Receivers in each treatment playing accept-reject (their wary equilibrium strategy) in each round. Bars indicate 95% confidence intervals.

Figure C.4
Percent of Strategic Senders (By Round and Treatment)



Notes. This figure displays the percent of strategic Senders in each round of each treatment. A Sender is categorized as “strategic” in round x if they play their optimal strategy (partial truth-telling) in round x and all subsequent rounds.